

Work Package B: the smoothed triple identity and its uniform error budget

Project proof draft

October 2, 2026

Task: proof and bookkeeping. Assumptions: **UNCONDITIONAL**. Ledger status: **proved-draft**. This draft develops the exact smoothed identity, its uniform budget, the interior vanishing theorem, the one-sided positive-quadrant boundary limit, and a native weighted Schwartz correlation theorem on an open hexagon. An explicit rational kernel profile and a summable height-localization error preserve that theorem's main term, identified below as three halves of the all-ordered sine benchmark. A self-contained theorem in the final section assembles these results in unit benchmark normalization. Global constant-kernel replacement and horizontal extraction remain separate Work Package B tasks.

1 Definitions and imported inputs

Let $\mathcal{I}$ index all occurrences of nontrivial zeros, with multiplicity and both signs of the ordinate. Write $\rho_j = 1/2 + \delta_j + i\gamma_j$, so $|\delta_j| < 1/2$. Equality of indices, complex zeros, and ordinates are three different conditions. All powers of positive real numbers use the real logarithm. For $x \geq 1$ and $t \in \mathbb{R}$, put

$$D_j(t) = 1 + ((t - \gamma_j) + i\delta_j)^2, \quad \mathcal{S}(x, t) = \sum_{j \in \mathcal{I}} \frac{2x^{\delta_j + i(\gamma_j - t)}}{D_j(t)}, \quad (1)$$

$$a_x(n) = \frac{\Lambda(n)}{\sqrt{n}} \min\{n/x, x/n\}, \quad \mathcal{P}(x, t) = \sum_{n \geq 1} a_x(n) e^{-it \log n}. \quad (2)$$

Here $\Lambda(1) = 0$. The three direct dependencies, all reconstructed in `proofs/pair_baseline.tex`, are:

- **PAIR-EF-CONVERGENCE-001** (label `lem:ef-convergence`): absolute and locally uniform convergence of both series;
- **PAIR-COUNT-KERNEL-001** (label `lem:count-kernel`):

$$Q(t) := \sum_j (1 + (t - \gamma_j)^2)^{-1} \leq C_0 \log(|t| + 2),$$

with an effective absolute constant C_0 ;

- **PAIR-EF-001** (label `lem:pair-ef`): the exact identity

$$\mathcal{S}(x, t) = -\mathcal{P}(x, t) + A_x(t) + H_x(t), \quad A_x(t) = x^{-1} \log(|t| + 2). \quad (3)$$

For precision, H_x is the following exact function, not an unspecified error symbol. With $s = 1/2 + it$ and $\chi(w) = 2^w \pi^{w-1} \sin(\pi w/2) \Gamma(1-w)$, set

$$\begin{aligned}
H_x &= E_{\text{arch}}(x, \cdot) + E_{\text{DS}}(x, \cdot) + E_{\text{pole}}(x, \cdot) + E_{\text{trivial}}(x, \cdot), \\
E_{\text{arch}}(x, t) &= x^{-1} \left[-\frac{\chi'}{\chi}(-1/2 + it) - \log(|t| + 2) \right], \\
E_{\text{DS}}(x, t) &= x^{-1} \frac{\zeta'}{\zeta}(3/2 - it), \\
E_{\text{pole}}(x, t) &= \frac{2x^{1/2-it}}{(1/2 + it)(3/2 - it)}, \\
E_{\text{trivial}}(x, t) &= 2 \sum_{m \geq 1} \frac{x^{-2m-s}}{(2m + s - 1)(2m + s + 1)}. \tag{4}
\end{aligned}$$

The last series is absolutely locally uniformly convergent, including $x = 1$. The imported separate bounds, uniform in $x \geq 1, t \in \mathbb{R}$, are

$$\begin{aligned}
|E_{\text{arch}}| &\leq C_{\text{arch}}/x, & |E_{\text{DS}}| &\leq C_{\text{DS}}/x, \\
|E_{\text{pole}}| &\leq 4\sqrt{x}/(1 + t^2), & |E_{\text{trivial}}| &\leq 12x^{-5/2}/(|t| + 2). \tag{5}
\end{aligned}$$

All constants here are effective. Their propagation through products will be a separate error-budget step.

2 The exact identity

Lemma 1 (TRIPLE-MASTER-001). *Fix a real nonnegative $\omega \in C_c^\infty((1, 2))$ with $\int_{\mathbb{R}} \omega(u) du = 1$. For $T \geq 3, X, Y \geq 1$, set $Z = XY$ and*

$$\mathfrak{I}_T(f, g, h) = \frac{1}{T} \int_{\mathbb{R}} \omega(t/T) f(t) g(t) \overline{h(t)} dt, \tag{6}$$

$$\mathcal{C}_{3,T}(X, Y) = \mathfrak{I}_T(\mathcal{S}(X, \cdot), \mathcal{S}(Y, \cdot), \mathcal{S}(Z, \cdot)),$$

$$K_T(j, k, \ell) = \frac{1}{T} \int_{\mathbb{R}} \frac{\omega(t/T)}{D_j(t) D_k(t) \overline{D_\ell(t)}} dt. \tag{7}$$

Then the following absolutely convergent expansion holds:

$$\mathcal{C}_{3,T}(X, Y) = 8 \sum_{j, k, \ell \in \mathcal{I}} X^{\rho_j + \overline{\rho_\ell} - 1} Y^{\rho_k + \overline{\rho_\ell} - 1} K_T(j, k, \ell). \tag{8}$$

Its five disjoint index subseries are listed in Table 1. In particular $\mathcal{C}_{3,T}(X, Y) = \mathcal{C}_{3,T}(Y, X)$, and

$$|\mathcal{C}_{3,T}(X, Y)| \leq (8/3)^3 C_0^3 XY \log^3(2T + 2) \|\omega\|_1. \tag{9}$$

Write $P_x = \mathcal{P}(x, \cdot)$. For a word $w \in \{P, A, H\}^3$, let $N_P(w)$ be its number of letters P . There is also the exact expansion

$$\mathcal{C}_{3,T}(X, Y) = \sum_{w \in \{P, A, H\}^3} (-1)^{N_P(w)} \mathfrak{I}_T(w_{1X}, w_{2Y}, w_{3Z}). \tag{10}$$

Table 2 records all 27 ordered terms. In the Fourier convention $\widehat{\omega}(\xi) = \int \omega(u) e^{-2\pi i u \xi} du$, the unsigned cubic prime term is

$$\mathfrak{I}_T(P_X, P_Y, P_Z) = \sum_{m,n,k \geq 1} a_X(m) a_Y(n) a_Z(k) \widehat{\omega}\left(\frac{T}{2\pi} \log \frac{mn}{k}\right). \quad (11)$$

The series is absolutely convergent. Its exact resonance $mn = k$ equals

$$R_{\text{res}}(X, Y) = \sum_{p \text{ prime}} \sum_{a,b \geq 1} a_X(p^a) a_Y(p^b) a_Z(p^{a+b}). \quad (12)$$

If $R_{\text{off}}(X, Y, T)$ denotes the subseries in (11) with $mn \neq k$, its contribution together with the resonance to $\mathcal{C}_{3,T}$ is $-R_{\text{res}} - R_{\text{off}}$. No estimate making R_{off} negligible is asserted.

Proof. For real v , $|\delta| < 1/2$,

$$|1 + (v + i\delta)^2| \geq 1 + v^2 - \delta^2 \geq \frac{3}{4}(1 + v^2).$$

Consequently the sum of absolute values of the individual products in the integrand defining $\mathcal{C}_{3,T}$ is at most

$$(8/3)^3 (XYZ)^{1/2} Q(t)^3 \leq (8/3)^3 XY C_0^3 \log^3(|t| + 2).$$

Multiplication by $|\omega(t/T)|/T$ gives an integrable dominating function supported inside $(T, 2T)$. Tonelli applied to absolute values and then Fubini justify the three zero sums and their integration. They give (9) and absolute convergence even before integrating the absolute values of individual terms. The numerator of a product is

$$8X^{\delta_j + i(\gamma_j - t)} Y^{\delta_k + i(\gamma_k - t)} Z^{\delta_\ell - i(\gamma_\ell - t)} = 8X^{\delta_j + \delta_\ell + i(\gamma_j - \gamma_\ell)} Y^{\delta_k + \delta_\ell + i(\gamma_k - \gamma_\ell)}.$$

This uses $\log Z = \log X + \log Y$; all center phases cancel. The conjugated third denominator is $\overline{D_\ell(t)}$. This proves (8). Exchanging the first two slots proves the stated symmetry. Partitioning ordered index triples into their five equality patterns proves Table 1; absolute convergence allows this regrouping without a limiting-order convention.

Substitute (3) into the three slots of $\mathfrak{I}_T$. Each slot has three summands. The sign of a term is $(-1)^{N_P(w)}$, including when the third slot is $-P_Z$, because -1 is real. This proves (10) and Table 2. Every function is continuous on the compact integration window, so this finite distribution is legitimate. Each H may in turn be expanded into its four exact named terms in (4); conjugation remains on every third-slot term.

For the prime expansion, set $B(x) = \sum_{n \geq 1} a_x(n)$. The majorant $a_x(n) \leq x(\log n)n^{-3/2}$, $n \geq 2$, gives $B(x) < \infty$. The integral of the absolute product series is at most $\|\omega\|_1 B(X)B(Y)B(Z)$, which proves Fubini's hypotheses. The oscillation is $e^{-it(\log m + \log n - \log k)}$. Substituting $t = Tu$ cancels $1/T$ and gives exactly the argument $T \log(mn/k)/(2\pi)$ in (11). In particular there is no extra factor of T or 2π . Nonzero resonant coefficients require m, n, k all to be prime powers greater than one. Unique factorization and $mn = k$ then force $m = p^a, n = p^b, k = p^{a+b}$ for a single prime and $a, b \geq 1$. Conversely every such triple is resonant. Since $\widehat{\omega}(0) = 1$, (12) follows, with ordered (a, b) and no division by two. Both subseries converge absolutely by the preceding majorant. The negative sign is the *PPP* row of Table 2. $\square$

Partition	Condition	Contribution	Count
123	$j = k = \ell$	$\sum_j B_{jjj}$	q
12 3	$j = k \neq \ell$	$\sum_{j \neq \ell} B_{jj\ell}$	$q(q-1)$
13 2	$j = \ell \neq k$	$\sum_{j \neq k} B_{jkj}$	$q(q-1)$
23 1	$k = \ell \neq j$	$\sum_{j \neq k} B_{jkk}$	$q(q-1)$
1 2 3	all indices distinct	$\sum_{j,k,\ell \text{ distinct}} B_{jk\ell}$	$q(q-1)(q-2)$

Table 1: Disjoint index diagonals.

3 Zero-index bookkeeping

Set $B_{jk\ell} = 8X^{\rho_j + \bar{\rho}_\ell - 1}Y^{\rho_k + \bar{\rho}_\ell - 1}K_T(j, k, \ell)$. The last column gives counts for a finite set of q occurrences.

Distinct indices can represent repeated copies of the same complex zero; distinct complex zeros can have the same ordinate. No row imposes either of these stronger distinctions. The three partial diagonals have different kernels and weights; they have not been replaced by three times one row.

Horizontal reflection $\rho \mapsto 1 - \bar{\rho}$ acts by $(\delta, \gamma) \mapsto (-\delta, \gamma)$ and preserves multiplicity. Choose any corresponding bijection on occurrences. Simultaneous reindexing of all three slots preserves each row and its sum, by absolute convergence. In the reindexed summand the horizontal factor becomes $X^{-\delta_j - \delta_\ell}Y^{-\delta_k - \delta_\ell}$ and the denominator inside the kernel becomes $\overline{D_j} \overline{D_k} D_\ell$; the ordinate phases are unchanged. This is an equality of full subseries, not a termwise invariance. Reindexing only one slot preserves the unrestricted sum but need not preserve an index diagonal. Thus retaining these horizontal factors is compatible with functional-equation symmetry without setting any δ to zero.

4 Prime, archimedean, and remainder bookkeeping

In Table 2, each row denotes its sign times the normalized integral of the displayed product against $\omega(t/T)$. All factors are evaluated at t , and $Z = XY$. There are eight rows without H and nineteen with at least one H . The latter are retained exactly; their number is not an error estimate.

Table 2: The 27 ordered words in (10).

Word	Sign	Integrand	Contains H
PPP	−	$P_X P_Y \overline{P_Z}$	no
PPA	+	$P_X P_Y \overline{A_Z}$	no
PPH	+	$P_X P_Y \overline{H_Z}$	yes
PAP	+	$P_X A_Y \overline{P_Z}$	no
PAA	−	$P_X A_Y \overline{A_Z}$	no
PAH	−	$P_X A_Y \overline{H_Z}$	yes
PHP	+	$P_X H_Y \overline{P_Z}$	yes
PHA	−	$P_X H_Y \overline{A_Z}$	yes
PHH	−	$P_X H_Y \overline{H_Z}$	yes
APP	+	$A_X P_Y \overline{P_Z}$	no
APA	−	$A_X P_Y \overline{A_Z}$	no
APH	−	$A_X P_Y \overline{H_Z}$	yes
AAP	−	$A_X A_Y \overline{P_Z}$	no

Word	Sign	Integrand	Contains H
AAA	+	$A_X A_Y \overline{A_Z}$	no
AAH	+	$A_X A_Y \overline{H_Z}$	yes
AHP	−	$A_X H_Y \overline{P_Z}$	yes
AHA	+	$A_X H_Y \overline{A_Z}$	yes
AHH	+	$A_X H_Y \overline{H_Z}$	yes
HPP	+	$H_X P_Y \overline{P_Z}$	yes
HPA	−	$H_X P_Y \overline{A_Z}$	yes
HPH	−	$H_X P_Y \overline{H_Z}$	yes
HAP	−	$H_X A_Y \overline{P_Z}$	yes
HAA	+	$H_X A_Y \overline{A_Z}$	yes
HAH	+	$H_X A_Y \overline{H_Z}$	yes
HHP	−	$H_X H_Y \overline{P_Z}$	yes
HHA	+	$H_X H_Y \overline{A_Z}$	yes
HHH	+	$H_X H_Y \overline{H_Z}$	yes

5 Endpoints, uniformity, and the next estimates

The endpoints $X = 1$, $Y = 1$, and $T = 3$ are included. The smooth weight vanishes in neighborhoods of $t = T, 2T$; there are no half weights. The height window is a condition on the center t , not on the zeros: all ordinates, including $0 < |\gamma| < 3$ if present, remain in the sums. No use of L_T^{-1} or a change of local-spacing normalization is made.

Limit order is explicit: fix T, X, Y, ω ; truncate each zero sum at $|\gamma| \leq U_r$, including endpoints with full multiplicity; then let the three U_r tend to infinity independently. The displayed dominating function proves convergence in any order or jointly. Prime truncations are removed in the same way using $B(X)B(Y)B(Z)$. No $T \rightarrow \infty$ limit, moving Fourier support, or shrinking smoothing width has been interchanged with those limits. The bounds are locally uniform in the finite parameters and explicitly retain $\|\omega\|_1$.

Integration by parts, with no boundary terms, gives for every integer $N \geq 1$

$$|\widehat{\omega}(\xi)| \leq \|\omega\|_1, \quad |\widehat{\omega}(\xi)| \leq \frac{\|\omega^{(N)}\|_1}{(2\pi|\xi|)^N} \quad (\xi \neq 0). \quad (13)$$

For $mn \neq k$, the second denominator becomes $(T|\log(mn/k)|)^N$. Uniformity for a varying family of weights requires bounds on these derivative norms. Smooth compact height support does not supply a compact Fourier support or exact off-resonance removal. Near $mn = k$ a new arithmetic estimate is still needed.

As an algebraic lower-level check, replacing the second input of $\mathfrak{I}_T$ by the constant function 1 gives the smoothed pair form $T^{-1} \int \omega(t/T) f(t) \overline{h(t)} dt$. Taking $Y = 1$ in the actual lemma does *not* perform that replacement: $\mathcal{S}(1, t)$ remains a zero sum.

The following sections estimate the eight rows without H and propagate the four separate bounds in (5) through the other nineteen rows, before choosing any growing range for X, Y . The zero index diagonals and prime resonances are separate decompositions, with no asserted row-wise correspondence. A Fourier-support region, an asymptotic main term, positivity, and a new horizontal conclusion are not established by this lemma.

6 Uniform norm bounds and the named budget

The task remains a proof under UNCONDITIONAL assumptions. In addition to Lemma 1, the direct analytic inputs are PAIR-EF-001, PAIR-PRIME-MEAN-001, and PAIR-PRIME-DIAGONAL-001, in the pair draft at labels `lem:pair-ef`, `lem:prime-mean`, and `lem:prime-diagonal`. The latter two give, with effective absolute constants C_{pair}, C_D ,

$$\int_0^V |P_x(t)|^2 dt \leq C_{\text{pair}} V \log(2x) \quad (V \geq 3, 1 \leq x \leq V), \quad \sum_n a_x(n)^2 \leq C_D \log(2x) \quad (x \geq 1).$$

These are unconditional estimates including prime powers.

Keep the hypotheses on T, X, Y, Z, ω from Lemma 1, and define

$$\begin{aligned} d\mu_T(t) &= T^{-1} \omega(t/T) dt, & q &= \log T, & L &= \log(2T + 2), & \ell_x &= \log(2x), \\ W_0 &= \|\omega\|_1 = 1, & W_\infty &= \|\omega\|_\infty, & W_1 &= \|\omega'\|_1. \end{aligned} \quad (14)$$

The measure μ_T is a probability measure. Set

$$\begin{aligned} C_B &= 1 + \frac{2}{(1 - 2^{-1/2})^2}, & C_1 &= 1 + \frac{2}{1 - 2^{-1/2}}, & C_M &= \sqrt{2C_{\text{pair}}}, \\ b_x &= C_B \sqrt{x} \ell_x, & \alpha_x &= L/x, \\ m_x &= \begin{cases} \min\{b_x, C_M \sqrt{W_\infty \ell_x}\}, & 1 \leq x \leq 2T, \\ b_x, & x > 2T, \end{cases} \\ h_{\text{arch}}(x) &= C_{\text{arch}}/x, & h_{\text{DS}}(x) &= C_{\text{DS}}/x, \\ h_{\text{pole}}(x) &= \frac{4\sqrt{x}}{1 + T^2}, & h_{\text{trivial}}(x) &= \frac{12x^{-5/2}}{T + 2}, & h_x &= \sum_{\nu \in \mathcal{J}} h_\nu(x), \end{aligned} \quad (15)$$

where $\mathcal{J} = \{\text{arch}, \text{DS}, \text{pole}, \text{trivial}\}$. The subscripts x on m, h, α suppress their displayed dependence on T, ω .

For $r = 1, 2, 3$, put

$$J_r = \int \omega(u) \log^r(Tu + 2) du, \quad D(a, b) = \sum_{n \geq 1} a_a(n) a_b(n), \quad d_{a,b} = C_D \sqrt{\ell_a \ell_b}.$$

Integrals here and below involving $\log(Tu + 2)$ are over $(1, 2)$; the weighted functions are extended by zero outside that interval. In particular they are smooth and compactly supported. For $r = 1, 2$, set

$$\Omega_{r,1} = \|(\omega(u) \log^r(Tu + 2))'\|_1, \quad V_r = W_1 L^r + r W_0 L^{r-1}. \quad (17)$$

The notation $D(a, b)$ refers to prime coefficients and has no relation to the zero denominators $D_j(t)$.

Lemma 2 (TRIPLE-UNIFORM-001). *Uniformly for $T \geq 3$, $X, Y \geq 1$ and every admissible ω , the following bounds hold, with the smoothing dependence explicitly retained:*

$$\begin{aligned} \|P_x\|_\infty &\leq \sum_n a_x(n) \leq b_x, & \sum_{n \geq 2} \frac{a_x(n)}{\log n} &\leq C_1 \sqrt{x}, \\ \|P_x\|_{L^2(\mu_T)} &\leq m_x, & \|A_x\|_{L^\infty(T, 2T)} &\leq \alpha_x, & \|E_\nu(x, \cdot)\|_{L^\infty(T, 2T)} &\leq h_\nu(x), \\ 0 \leq D(a, b) &\leq d_{a,b}, & 0 \leq J_r &\leq L^r, & \Omega_{r,1} &\leq V_r \quad (r = 1, 2). \end{aligned} \quad (18)$$

Define the retained expression

$$M_T = \frac{q^3}{Z^2} + \frac{J_1}{Y} D(X, Z) + \frac{J_1}{X} D(Y, Z) - R_{\text{res}}(X, Y). \quad (19)$$

It is not asserted to be an asymptotic main term. There is the exact identity

$$\begin{aligned} \mathcal{C}_{3,T} = & M_T + E_{\text{archcube}} + E_{PPA} + E_{PAA} + E_{APA} + E_{AAP} \\ & + E_{PAP,\text{off}} + E_{APP,\text{off}} + E_{PPP,\text{off}} + \sum_{\substack{w \in \{P,A,H\}^3 \\ H \text{ occurs in } w}} E_w. \end{aligned} \quad (20)$$

The eight errors on the first two lines are defined and bounded in Table 3. In that table,

$$U = \min\{b_X m_Y m_Z, b_Y m_X m_Z, b_Z m_X m_Y\}, \quad 0 \leq R_{\text{res}} \leq b_X b_Y b_Z, \quad (21)$$

and the complete cubic satisfies $|\mathfrak{I}_T(P_X, P_Y, P_Z)| \leq U$.

For the remaining nineteen words, let $x_1 = X, x_2 = Y, x_3 = Z$, and $S_H(w) = \{i : w_i = H\}$. For each assignment $\kappa : S_H(w) \rightarrow \mathcal{J}$, replace the corresponding factors by $E_{\kappa(i)}(x_i, \cdot)$, retaining P, A elsewhere, to get functions f_1, f_2, f_3 . Define

$$E_{w;\kappa} = (-1)^{N_P(w)} \mathfrak{I}_T(f_1, f_2, f_3), \quad E_w = \sum_{\kappa} E_{w;\kappa}. \quad (22)$$

Then, separately for every assignment,

$$|E_{w;\kappa}| \leq \prod_{\{i:w_i=P\}} m_{x_i} \prod_{\{i:w_i=A\}} \alpha_{x_i} \prod_{\{i:w_i=H\}} h_{\kappa(i)}(x_i). \quad (23)$$

Summing these nonnegative bounds gives

$$|E_w| \leq \prod_{\{i:w_i=P\}} m_{x_i} \prod_{\{i:w_i=A\}} \alpha_{x_i} \prod_{\{i:w_i=H\}} h_{x_i}. \quad (24)$$

There are 208 refined remainder terms, grouped into nineteen words; the refined terms and their aggregates are two descriptions of the same summands. All constants in these assertions are effective and independent of T, X, Y, ω , apart from the displayed norms.

Proof. For $n \leq x$, the inequality $\Lambda(n) \leq \log n$ gives

$$\sum_{2 \leq n \leq x} a_x(n) \leq x^{-1} \sum_{2 \leq n \leq x} \sqrt{n} \log n \leq \sqrt{x} \ell_x.$$

For $j \geq 0$, the interval $2^j x < n \leq 2^{j+1} x$ contains at most $2^{j+1} x$ positive integers. Consequently its contribution is at most

$$2\sqrt{x} 2^{-j/2} \log(2^{j+1} x) \leq 2\sqrt{x} \ell_x (j+1) 2^{-j/2}.$$

Summing the geometric series and its derivative gives b_x as stated. Using $\Lambda(n)/\log n \leq 1$ in the same argument removes the factor $(j+1)\ell_x$, giving $C_1 \sqrt{x}$. The split counts an integer $n = x$ exactly once and applies also to $1 \leq x < 2$. All series are nonnegative and convergent.

Since μ_T has mass one, $\|P_x\|_{L^2(\mu_T)} \leq b_x$. If $x \leq 2T$, positivity of the weight and the imported mean-square bound with $V = 2T$ give

$$\int |P_x|^2 d\mu_T \leq \frac{W_\infty}{T} \int_0^{2T} |P_x(t)|^2 dt \leq 2C_{\text{pair}} W_\infty \ell_x.$$

This proves the piecewise bound, including $x = 2T$. No extension of the imported mean-square theorem to $x > 2T$ is used. The A and four remainder bounds follow pointwise from (5) on $T < t < 2T$. Cauchy–Schwarz for the coefficient sequences and the imported diagonal bound give $D(a, b) \leq \sqrt{D(a, a)D(b, b)} \leq d_{a, b}$.

On the support of ω , $q \leq \log(Tu + 2) \leq L$ and $T/(Tu + 2) \leq 1$. Thus $q^r \leq J_r \leq L^r$, and the product rule gives

$$\Omega_{r,1} \leq \|\omega'\|_1 L^r + r\|\omega\|_1 L^{r-1} = V_r.$$

There are no endpoint derivative contributions: ω vanishes in neighborhoods of 1, 2. In particular $0 \leq E_{\text{archcube}} \leq (L^3 - q^3)/Z^2$.

For the oscillating terms, set

$$F_r(\lambda) = \int \omega(u) \log^r(Tu + 2) e^{-iTu\lambda} du.$$

One integration by parts gives

$$|F_r(\lambda)| \leq \frac{\Omega_{r,1}}{T|\lambda|} \quad (\lambda \neq 0). \quad (25)$$

In project Fourier units $F_r(\lambda)$ is the transform of $\omega(u) \log^r(Tu + 2)$ at $T\lambda/(2\pi)$. The 2π from differentiation cancels against this denominator; the normalized substitution $t = Tu$ introduces no further factor. Since all prime coefficient sums are absolutely convergent and the weights bounded, Fubini applies at fixed parameters.

The *PPA* integral is

$$\frac{1}{Z} \sum_{m,n \geq 2} a_X(m) a_Y(n) F_1(\log(mn)).$$

Here $\log(mn) \geq \log 4$, which proves its table bound. The *PAA*, *APA*, *AAP* integrals have respective frequencies $\log n, \log n, -\log n$ and scalar denominators YZ, XZ, XY . Apply (25) and the bound for $\sum a_x(n)/\log n$. The third of these terms still has its conjugation; the change of frequency sign does not affect the absolute bound.

For example the mixed term is exactly

$$\mathfrak{J}_T(P_X, A_Y, P_Z) = \frac{1}{Y} \sum_{m,k \geq 2} a_X(m) a_Z(k) F_1(\log(m/k)).$$

The diagonal is $J_1 D(X, Z)/Y$. Cauchy–Schwarz bounds the full integral in absolute value by $(L/Y)m_X m_Z$. Subtracting its diagonal gives the bound in the table by the triangle inequality. The *APP* calculation is identical with X, Y interchanged. These bounds do not treat the near frequencies as separated from zero.

For the cubic integral take any one factor in sup norm and the other two in $L^2(\mu_T)$. The three choices give U . The resonant sum is a nonnegative subseries of the product of the three coefficient sums, so $R_{\text{res}} \leq b_X b_Y b_Z$. Its signed off-resonance error is $-R_{\text{off}}$ from Lemma 1; the triangle inequality gives $U + R_{\text{res}}$.

Each word containing H has at most two P slots. Bound all other factors in sup norm, and use Cauchy–Schwarz for two prime factors or $\|P_x\|_{L^1(\mu_T)} \leq \|P_x\|_{L^2(\mu_T)}$ for one. This proves (23). Finite distributivity proves (24), with no absorbed constants. There are twelve words with one H , six with two, and one with three:

$$12 \cdot 4 + 6 \cdot 4^2 + 4^3 = 208.$$

Finally substitute the exact signed definitions into (10). The *AAA* term is $J_3/(XYZ) = J_3/Z^2$. Extracting its q^3/Z^2 , the two mixed diagonals, and the negatively signed cubic resonance proves (20). $\square$

7 Range, interpretation, and remaining losses

All estimates in Lemma 2 hold on the entire range $T \geq 3$, $X, Y \geq 1$, using the specified branch of m_x . If $XY \leq 2T$, all three prime factors admit the sharper branch since $1 \leq X, Y \leq XY$; equality $XY = 2T$ is included. The line $x = 2T$ is a boundary of an estimate's applicability, not a Fourier-support boundary. For $x > 2T$ the elementary bound remains valid. The norm majorant need not be continuous when its branch changes.

Every zero occurrence from Lemma 1 is still retained. Its symmetry under simultaneous horizontal reflection therefore persists; estimating its exactly equal prime-side expression does not set $\delta_\rho = 0$. Swapping X, Y interchanges PAA, APA and the two mixed off-diagonal errors; the other definitions transform by their ordered slots. Setting a slot function to 1 recovers the ordinary weighted pair Cauchy–Schwarz bound. Setting its base to 1 does not remove that slot.

The limit order remains fixed T, X, Y, ω , followed by removal of absolutely convergent prime or zero cutoffs. Bounds involving ω' are used only after the weighted integrals are defined; no differentiation of an infinite prime series is taken. All displayed estimates are then uniform on their stated ranges. No shrinking-width limit is taken. Varying weights must retain W_∞, W_1 ; mass normalization alone does not bound them.

The pure archimedean term and the one-prime and same-sign two-prime terms have elementary bounds, including the displayed $1/T$ oscillatory gain. The mixed-prime diagonal bound imports PNT-level information through **PAIR-PRIME-DIAGONAL-001**. The weighted prime norms import the unconditional mean-value and prime-counting inputs in **PAIR-PRIME-MEAN-001**. No new prime-pair or prime-triple conjecture is used. The mixed-prime and cubic off-diagonal bounds in this lemma are triangle-inequality bounds and do not yield a uniform negligible error. No rowwise matching with zero-index diagonals has been established.

The file `research/triple-error-budget.md` records the nineteen remainder-containing rows and their component assignments. They form one budget: an aggregate error and its component terms must never both be added. There is no height-truncation, desmoothing, or support-boundary error in this identity. The following sections sharpen the mixed-prime and cubic off-diagonal bounds and combine the budget on a growing parameter range. The bounds of Lemma 2 alone do not settle an asymptotic or supply horizontal information.

8 An absolute logarithmic-gap estimate

The next three claims have assumptions **UNCONDITIONAL** and status **proved-draft**. The integer-frequency estimate is proved here; no additional mean-value or prime-correlation theorem is imported.

Lemma 3 (**TRIPLE-LOG-GAP-001**). *For complex sequences $c = (c_n)_{n \geq 2}$, $d = (d_n)_{n \geq 2}$, define*

$$B(c) = \sum_{n \geq 2} |c_n|, \quad \mathcal{E}(c) = \sum_{n \geq 2} n \log(2n) |c_n|^2,$$

and likewise for d . If these four quantities are finite, then

$$\sum_{\substack{m, n \geq 2 \\ m \neq n}} \frac{|c_m d_n|}{|\log(m/n)|} \leq \frac{B(c)B(d)}{\log 2} + 4\sqrt{2} \sqrt{\mathcal{E}(c)\mathcal{E}(d)}. \quad (26)$$

The sum is over ordered, distinct integer frequencies; repeated frequencies must be combined before applying the lemma.

Proof. Start with finitely supported sequences. For $n < m/2$ or $n > 2m$, the reciprocal logarithm is at most $1/\log 2$. This part is bounded by the first term on the right. On the complementary comparable range $m/2 \leq n \leq 2m$, $m \neq n$, integration of $1/u$ between m, n gives

$$|\log(m/n)| \geq \frac{|m-n|}{\max(m, n)}, \quad \max(m, n) \leq \sqrt{2mn}.$$

Set $u_m = \sqrt{m}|c_m|$, $v_n = \sqrt{n}|d_n|$. The comparable part is at most

$$\sqrt{2} \sum_{\substack{m/2 \leq n \leq 2m \\ m \neq n}} \frac{u_m v_n}{|m-n|} \leq \sqrt{2} \left(\sum_{\substack{m/2 \leq n \leq 2m \\ m \neq n}} \frac{u_m^2}{|m-n|} \right)^{1/2} \left(\sum_{\substack{m/2 \leq n \leq 2m \\ m \neq n}} \frac{v_n^2}{|m-n|} \right)^{1/2}.$$

For fixed $m \geq 2$, the row sum of $1/|m-n|$ is at most $2 \sum_{h=1}^{2m} 1/h \leq 2(1 + \log(2m)) \leq 4 \log(2m)$. The comparable relation is symmetric, so the same bound holds for column sums. This proves the second term of (26). The equal-ratio boundaries $n = 2m$, $m = 2n$ belong to the comparable part; $m = n$ is always excluded. Apply the finite result to independent initial truncations and pass to the limit by monotone convergence of nonnegative sums. The right side is finite, proving absolute convergence. $\square$

9 Sharper prime off-diagonal bounds

Lemma 4 (TRIPLE-OFFDIAG-001). *With the hypotheses and signed errors of Lemma 2, uniformly for every $T \geq 3$, $X, Y \geq 1$, $Z = XY$,*

$$|E_{PPP, \text{off}}| \ll W_1 \frac{Z \ell_Z^4}{T}, \quad (27)$$

$$|E_{PAP, \text{off}}| \ll \frac{\Omega_{1,1}}{T} \frac{X}{\sqrt{Y}} (\ell_X \ell_Z)^{3/2}, \quad (28)$$

$$|E_{APP, \text{off}}| \ll \frac{\Omega_{1,1}}{T} \frac{Y}{\sqrt{X}} (\ell_Y \ell_Z)^{3/2}.$$

Here $\Omega_{1,1} \leq W_1 L + 1$. The constants are absolute and effective, independent of smoothing apart from the displayed norms. No restriction $Z \leq 2T$ is needed for these estimates.

Proof. Write $v_x(n) = \min(n/x, x/n)$ and group the product series as

$$c_{X,Y}(r) = \sum_{\substack{mn=r \\ m, n \geq 2}} a_X(m) a_Y(n), \quad P_X(t) P_Y(t) = \sum_{r \geq 2} c_{X,Y}(r) r^{-it}. \quad (29)$$

Each coefficient is a finite sum over ordered factorizations. Absolute convergence from Lemma 2, or Tonelli for the nonnegative coefficients, gives

$$B(c_{X,Y}) = B(a_X) B(a_Y) \leq C_B^2 \sqrt{Z} \ell_X \ell_Y.$$

Thus repeated product frequencies have been combined with their full coefficient, not discarded.

Since $v_x(n) = \exp(-|\log(n/x)|)$, the triangle inequality gives

$$v_X(m) v_Y(n) \leq v_Z(mn). \quad (30)$$

The elementary divisor identity $\sum_{d|r} \Lambda(d) = \log r$ follows by writing $r = \prod p^{a_p}$: each prime contributes $a_p \log p$. Since $0 \leq \Lambda(r/d) \leq \log r$,

$$(\Lambda * \Lambda)(r) = \sum_{d|r} \Lambda(d) \Lambda(r/d) \leq (\log r)^2.$$

Consequently

$$0 \leq c_{X,Y}(r) \leq \frac{v_Z(r)(\log r)^2}{\sqrt{r}}. \quad (31)$$

For $k = 3, 5$, elementary dyadic summation gives, for $x \geq 1$,

$$\sum_{n \geq 2} v_x(n)^2 \log^k(2n) \leq K_k x \ell_x^k, \quad K_k = 1 + 2 \sum_{j \geq 0} 2^{-j} (j+2)^k < \infty. \quad (32)$$

Indeed the part $n \leq x$ has at most x terms, each at most ℓ_x^k . On $2^j x < n \leq 2^{j+1} x$, use at most $2^{j+1} x$ terms, $v_x(n)^2 \leq 2^{-2j}$, and $\log(2n) \leq (j+2)\ell_x$. These disjoint intervals include upper endpoints with full weight. The constants are effective: geometric series and their first k derivatives evaluate the displayed sums. This proves, with the energy from Lemma 3,

$$\mathcal{E}(a_x) \leq K_3 x \ell_x^3, \quad \mathcal{E}(c_{X,Y}) \leq K_5 Z \ell_Z^5. \quad (33)$$

These arguments also cover $x = 1$, with an empty lower interval.

The cubic off-diagonal, with its original sign, is

$$E_{PPP,\text{off}} = - \sum_{\substack{r,k \geq 2 \\ r \neq k}} c_{X,Y}(r) a_Z(k) \int \omega(u) e^{-iTu \log(r/k)} du.$$

Grouping and integrating this series is justified by $B(c_{X,Y})B(a_Z) < \infty$. One integration by parts bounds each integral by $W_1/(T|\log(r/k)|)$. By Lemma 3, the sum of these bounds is finite and is

$$\ll \frac{W_1}{T} (Z \ell_X \ell_Y \ell_Z + Z \ell_Z^4) \ll \frac{W_1}{T} Z \ell_Z^4.$$

Here $\ell_X, \ell_Y \leq \ell_Z$ and $\ell_Z \geq \log 2$. The removed diagonal is exactly $\sum_r c_{X,Y}(r) a_Z(r) = R_{\text{res}}(X, Y)$: expanding $c_{X,Y}$ restores $mn = k$ and the ordered same-prime-power classification of Lemma 1. Its negative sign in the retained expression is unchanged.

For the mixed terms, the already established representation gives

$$E_{PAP,\text{off}} = \frac{1}{Y} \sum_{\substack{m,k \geq 2 \\ m \neq k}} a_X(m) a_Z(k) F_1(\log(m/k)).$$

Apply (25), the gap lemma, and (33). For arbitrary $a, b \geq 1$ the resulting coefficient bound is

$$B(a_a)B(a_b) + \sqrt{\mathcal{E}(a_a)\mathcal{E}(a_b)} \ll \sqrt{ab}(\ell_a \ell_b)^{3/2};$$

the first term is absorbed using $\sqrt{\ell_a \ell_b} \geq \log 2$. Taking $a = X, b = Z$, and using $\sqrt{XZ}/Y = X/\sqrt{Y}$, proves (28). Exchanging X, Y proves the other bound.

All integrations by parts act on compact smooth weights, with no boundary terms. The Fourier frequency is $T \log(r/k)/(2\pi)$; the 2π derivative factor cancels, and $dt/T = du$. In particular no factor of T or 2π is omitted. Parameters are fixed while coefficient cutoffs are removed. The absolutely summable reciprocal-log bounds justify that removal in any order; only afterwards are the uniform estimates used. $\square$

10 An additive formula on an interior growing range

Corollary 5 (TRIPLE-SMOOTHED-ADDITIVE-001). *Let $0 < \varepsilon < 1/3$, $T \geq 3$, and*

$$T^\varepsilon \leq X, Y, \quad XY \leq T^{1-\varepsilon}. \quad (34)$$

For every admissible ω , with M_T defined in (19), one has

$$|\mathcal{C}_{3,T}(X, Y) - M_T| \leq C(1 + W_\infty + W_1)T^{-\varepsilon} \log^4(2T + 2), \quad (35)$$

where C is absolute and effective, independent of $\varepsilon, T, X, Y, \omega$. For fixed ε, ω , this is $\mathcal{C}_{3,T}(X, Y) = M_T + o(1)$ uniformly over (34). It is an additive approximation for the present smoothed observable, not an assertion of relative error $o(M_T)$.

Proof. All three bases $x = X, Y, Z$ satisfy $T^\varepsilon \leq x \leq T^{1-\varepsilon} < 2T$, and $Z \geq T^{2\varepsilon}$. Thus $\ell_x \leq L$ and Lemma 2 gives

$$m_x \ll \sqrt{W_\infty L}, \quad \alpha_x \leq T^{-\varepsilon} L, \quad h_x \ll T^{-\varepsilon}.$$

For the last estimate the archimedean and Dirichlet-series components are bounded by constants times $1/x$. The pole bound is at most $4T^{-(3+\varepsilon)/2}$, and the trivial-zero bound is at most $12T^{-1-5\varepsilon/2}$; both are at most a constant times $T^{-\varepsilon}$ on the stated epsilon range.

There are three remainder-containing words with two P 's, nine with one, and seven with none. Their component bounds sum to

$$\begin{aligned} \sum_{w: H \text{ occurs}} |E_w| &\ll W_\infty T^{-\varepsilon} L + \sqrt{W_\infty} T^{-2\varepsilon} L^{3/2} + T^{-3\varepsilon} L^2 \\ &\ll (1 + W_\infty) T^{-\varepsilon} L^2. \end{aligned} \quad (36)$$

This uses aggregates once, not again alongside their 208 components.

The other errors from the exact budget have the following separate bounds on (34):

Error	Absolute bound up to an effective absolute constant
E_{archcube}	$T^{-4\varepsilon} L^2$
E_{PPA}	$(1 + W_1) T^{-1-\varepsilon} L^3$
E_{PAA}, E_{APA}	$(1 + W_1) T^{-1-5\varepsilon/2} L^2$
E_{AAP}	$(1 + W_1) T^{-1-\varepsilon} L^2$
$E_{PAP, \text{off}}, E_{APP, \text{off}}$	$(1 + W_1) T^{-\varepsilon} L^4$
$E_{PPP, \text{off}}$	$W_1 T^{-\varepsilon} L^4$

For the archimedean error use $L^3 - q^3 \leq 3(L - q)L^2$ and $L - q = \log(2 + 2/T) \leq \log(8/3)$. For PPA , use $b_X b_Y \ll \sqrt{Z} L^2$ and $\Omega_{1,1} \leq (1 + W_1)L$. The denominators of the one-prime errors simplify respectively to $T\sqrt{X}Y^2$, $T\sqrt{Y}X^2$, $T\sqrt{Z}$, while $\Omega_{2,1} \leq (W_1 + 2)L^2$. For the mixed errors use $X/\sqrt{Y} \leq Z$, $Y/\sqrt{X} \leq Z$, and $Z/T \leq T^{-\varepsilon}$; the cubic bound follows directly from (27).

Every row and (36) is bounded by the right side of (35), with a constant independent of epsilon. Summing the exact budget (20) proves the result. Finally $T^{-\varepsilon} \log^4(2T + 2) \rightarrow 0$ for each fixed positive epsilon, proving the uniform additive assertion for fixed smoothing. $\square$

The lower and upper parameter boundaries in (34) are included, as is $T = 3$. The range is nonempty for every allowed epsilon: $X = Y = T^\varepsilon$ is an example. No assertion is made at $\varepsilon = 0$, or for a varying epsilon tending to zero. The explicit bound (35) applies to varying weights, but its $o(1)$ consequence then requires the displayed product with $1 + W_\infty + W_1$ to tend to zero.

The limit order is fixed T, X, Y, ω , removal of the convergent coefficient and zero cutoffs, and then $T \rightarrow \infty$ with fixed ε, ω and arbitrary X, Y in the displayed range. All full-zero coordinates, multiplicities, and horizontal reflection conventions remain those of Lemma 1; no zero-side truncation or substitution of $\beta = 1/2$ has occurred. The formulas and bounds respect exchange of X, Y .

The parameter bounds above do not by themselves constitute a test-function theorem. The following sections evaluate the retained quantities and derive the precise weighted test-function consequence. No dominance over the additive error, GUE main term, general Work Package B theorem, or new horizontal consequence is asserted.

11 Evaluation of the retained expression

These claims remain UNCONDITIONAL, with status **proved-draft**. The additional analytic input is the effective classical PNT already registered as **PRIME-PNT-001**, in the pair draft at label **inp:prime-pnt**. No new literature theorem is imported.

Lemma 6 (TRIPLE-RETAINED-001). *For the existing normalized weight, set $\mu_\omega = \int_1^2 \omega(u) \log u \, du$. Then, for $T \geq 3$,*

$$0 \leq J_1 - \log T - \mu_\omega \leq 2/T, \quad 0 \leq \mu_\omega \leq \log 2. \quad (37)$$

Uniformly for $1 \leq a \leq b$, the full prime-power mixed diagonal is

$$D(a, b) = \frac{a}{2b} (1 + \log(b/a)) \log(ab) + O((a/b)(1 + \log(b/a))). \quad (38)$$

For reversed arguments use $D(a, b) = D(b, a)$. Uniformly for $X, Y \geq 1$,

$$0 \leq R_{\text{res}}(X, Y) \ll Z^{-1/2} \ell_Z^3, \quad Z = XY. \quad (39)$$

All constants are absolute and effective. Define

$$G(X, Y) = \frac{(1 + \log Y) \log(XZ)}{2Y^2} + \frac{(1 + \log X) \log(YZ)}{2X^2}. \quad (40)$$

Let $\Delta(a, b)$ be $D(a, b)$ minus the displayed leading expression in (38), with the arguments ordered first. Exactly,

$$\begin{aligned} M_T &= q^3/Z^2 + (q + \mu_\omega)G + E_D + E_J - R_{\text{res}}, \\ E_D &= J_1 \{ \Delta(X, Z)/Y + \Delta(Y, Z)/X \}, \\ E_J &= (J_1 - q - \mu_\omega)G, \\ |E_D| &\ll L \left\{ \frac{1 + \log Y}{Y^2} + \frac{1 + \log X}{X^2} \right\}, \quad 0 \leq E_J \leq 2G/T. \end{aligned} \quad (41)$$

This evaluation has additive errors; no relative precision is asserted.

Proof. On $1 < u < 2$, $\log(Tu + 2) = q + \log u + \log(1 + 2/(Tu))$, and the last term lies between zero and $2/T$. Integrating against the nonnegative unit-mass weight proves (37).

Let $N_{\text{pp}}(u)$ count distinct integers that are proper prime powers, and let it be zero below 2. Counting all integer powers gives, for $u \geq 2$,

$$N_{\text{pp}}(u) \leq \sqrt{u} + u^{1/3} \frac{\log u}{\log 2} \leq \left(1 + \frac{6}{e \log 2} \right) \sqrt{u}.$$

The last inequality follows by maximizing $ye^{-y/6}$ for $y \geq 0$. Overcounting integers with multiple power representations is harmless. PNT and partial summation give $\sum_{p \leq u} (\log p)^2 = u \log u + O(u)$. For example, use $\vartheta(u) \log u - \int_2^u \vartheta(v)/v dv$ and the existing effective bound for $\vartheta(u) - u$. The proper-power contribution is at most $N_{\text{pp}}(u) \log^2 u \ll \sqrt{u} \log^2 u \ll u$. Thus, including the bounded range $1 \leq u < 2$,

$$A(u) := \sum_{n \leq u} \Lambda(n)^2 = u \log u + E(u), \quad |E(u)| \leq Cu, \quad E(1) = 0.$$

For $1 \leq a \leq b$, the continuous weight for $D(a, b)$ is

$$w_{a,b}(u) = \begin{cases} u/(ab), & 1 \leq u \leq a, \\ a/(bu), & a < u \leq b, \\ ab/u^3, & u > b. \end{cases}$$

Stieltjes integration by parts gives

$$D(a, b) = \int_1^\infty w_{a,b}(u)(\log u + 1) du + O\left(\int_1^\infty u|w'_{a,b}(u)| du\right).$$

The lower error boundary vanishes because $E(1) = 0$; the upper one vanishes by the u^{-3} decay. Continuity at a, b removes all jump terms, including $a = b$ and integer prime-power endpoints. Writing $r = a/b$, $h = \log(b/a)$, direct integration yields

$$\begin{aligned} \int_1^\infty u|w'_{a,b}(u)| du &= r(2+h) - \frac{1}{2ab}, \\ \int_1^\infty w_{a,b}(u)(\log u + 1) du &= \frac{r}{2}(1+h) \log(ab) + r(1+h) - \frac{1}{4ab}. \end{aligned}$$

Since $1/(ab) \leq r$, this proves (38). The improper operations can first be made on finite intervals; the displayed majorants justify their limits at fixed a, b .

For the resonance, only $r = p^k$, $k \geq 2$, contributes to $\sum_r c_{X,Y}(r) a_Z(r)$. Its divisor convolution is exactly $(\Lambda * \Lambda)(p^k) = (k-1)(\log p)^2$. Using (30),

$$R_{\text{res}} \leq \sum_{\substack{r=p^k \\ k \geq 2}} \frac{v_Z(r)^2}{r} (k-1)(\log p)^3 \leq \sum_{\substack{r=p^k \\ k \geq 2}} \frac{v_Z(r)^2}{r} \log^3 r.$$

On $Z2^{-j-1} < r \leq Z2^{-j}$, $j \geq 0$, the last weight is at most $2^{-j}Z^{-1}\ell_Z^3$; the count is $O(\sqrt{Z}2^{-j/2})$. On $2^jZ < r \leq 2^{j+1}Z$, it is at most $2^{-3j}Z^{-1}(j+2)^3\ell_Z^3$, with count $O(\sqrt{Z}2^{j/2})$. Both geometric sums converge, proving (39), including $Z = 1$. These are disjoint ranges with full upper endpoints; the original ordered factorizations were already combined in $c_{X,Y}$. Finally substitute (38) and (37) into (19). Since $G \geq 0$, this proves all of (41). $\square$

Corollary 7 (TRIPLE-INTERIOR-VANISHING-001). *Uniformly for $0 < \varepsilon < 1/3$, $T \geq 3$ and $T^\varepsilon \leq X, Y, XY \leq T^{1-\varepsilon}$,*

$$\begin{aligned} |M_T| &\ll T^{-\varepsilon}L^3, \\ |\mathcal{C}_{3,T}(X, Y)| &\ll (1 + W_\infty + W_1)T^{-\varepsilon}L^4. \end{aligned} \tag{42}$$

The effective constants are absolute, including independence of epsilon. In particular $\mathcal{C}_{3,T} \rightarrow 0$ uniformly on this range for fixed positive epsilon and fixed smoothing.

Proof. Here $Z \geq T^{2\varepsilon}$ and $\ell_X, \ell_Y, \ell_Z \leq L$. Lemma 6 bounds the archimedean term by $T^{-4\varepsilon}L^3$, the two mixed contributions by $O(T^{-2\varepsilon}L^3)$, and the resonance by $O(T^{-\varepsilon}L^3)$. This proves the first bound; combine it with Corollary 5 for the second. All constants are uniform and $L > 1$. The asymptotic consequence fixes epsilon and smoothing before taking $T \rightarrow \infty$. $\square$

12 An interior weighted test-function theorem

Define

$$\Omega_+ = \{(\xi, \eta) : \xi > 0, \eta > 0, \xi + \eta < 1\}, \quad \Omega_\varepsilon = \{(\xi, \eta) : \xi, \eta \geq \varepsilon, \xi + \eta \leq 1 - \varepsilon\}.$$

For $0 < \varepsilon < 1/3$, this compact triangle is contained in Ω_+ . Every compact subset of Ω_+ is contained in some Ω_ε . This margin does not assert extension to the axes or to $\xi + \eta = 1$.

Theorem 8 (TRIPLE-TEST-FUNCTION-001). *Let $\phi \in C_c^\infty(\Omega_+)$ be complex-valued, and set*

$$F(z_1, z_2) = \int_{\mathbb{R}^2} \phi(\xi, \eta) e^{2\pi i(\xi z_1 + \eta z_2)} d\xi d\eta. \quad (43)$$

Then $F|_{\mathbb{R}^2}$ is Schwartz, $\widehat{F} = \phi$ in the project convention, and the displayed expression is its entire extension. For $T > 2\pi$, put $B = T/(2\pi)$, $L_T = \log B/(2\pi)$, and

$$\begin{aligned} z_{21} &= L_T \{(\gamma_{i_2} - \gamma_{i_1}) - i(\delta_{i_2} + \delta_{i_1})\}, \\ z_{31} &= L_T \{(\gamma_{i_3} - \gamma_{i_1}) - i(\delta_{i_3} + \delta_{i_1})\}. \end{aligned} \quad (44)$$

There is the exact identity of absolutely convergent expressions

$$\begin{aligned} \mathcal{R}_{3,T}^K[F; \omega] &:= 8 \sum_{i_1, i_2, i_3 \in \mathcal{I}} K_T(i_2, i_3, i_1) F(z_{21}, z_{31}) \\ &= \int_{\mathbb{R}^2} \phi(\xi, \eta) \mathcal{C}_{3,T}(B^\xi, B^\eta) d\xi d\eta. \end{aligned} \quad (45)$$

All ordered zero occurrences, both ordinate signs, full multiplicities, and all five index-diagonal patterns are included. There is no additional height or mean-spacing normalization outside the existing K_T . If $\text{supp } \phi \subset \Omega_\varepsilon$, then

$$|\mathcal{R}_{3,T}^K[F; \omega]| \ll \|\phi\|_1 (1 + W_\infty + W_1) B^{-\varepsilon} \log^4(2T + 2). \quad (46)$$

The constant is absolute and effective. For fixed $\phi, \varepsilon, \omega$, the weighted statistic tends to zero as $T \rightarrow \infty$.

Proof. Compact frequency support permits differentiation under (43), locally uniformly in complex z_1, z_2 , proving the entire extension. Repeated integration by parts in the frequency variables proves rapid decay of every real-variable derivative. Schwartz Fourier inversion, with the stated 2π convention, gives $\widehat{F} = \phi$.

Apply Lemma 1 with $X = B^\xi$, $Y = B^\eta$, and relabel $(j, k, \ell) = (i_2, i_3, i_1)$. The numerator is exactly

$$X^{\rho_{i_2} + \bar{\rho}_{i_1} - 1} Y^{\rho_{i_3} + \bar{\rho}_{i_1} - 1} = e^{2\pi i(\xi z_{21} + \eta z_{31})}.$$

In particular the imaginary shifts have a minus sign and involve sums of horizontal displacements. The third slot is the conjugated anchor in $K_T(i_2, i_3, i_1)$.

For fixed T and support in Ω_ε , the integrated absolute zero-series majorant from the master proof bounds

$$8 \sum_{i_1, i_2, i_3} \int_{\mathbb{R}^2} |\phi(\xi, \eta) K_T(i_2, i_3, i_1)| \left| e^{2\pi i(\xi z_{21} + \eta z_{31})} \right| d\xi d\eta \ll \|\phi\|_1 B^{1-\varepsilon} \log^3(2T+2).$$

Indeed $XY = B^{\xi+\eta} \leq B^{1-\varepsilon}$ on the support. This finite dominating quantity proves Tonelli/Fubini, the identity, and absolute convergence of the zero expression. No bound on F at complex arguments is inferred from real-axis Schwartz decay.

To use the precise spacing convention, set

$$\theta_T = \frac{\log B}{\log T} \in (0, 1), \quad \varepsilon_T = \varepsilon \theta_T \in (0, 1/3).$$

For $(\xi, \eta) \in \Omega_\varepsilon$, $X, Y \geq T^{\varepsilon_T}$, and

$$Z \leq T^{(1-\varepsilon)\theta_T} \leq T^{1-\varepsilon_T},$$

where the last inequality is just $\theta_T \leq 1$. Apply the explicit bound of Corollary 7 at this value of its margin, using its constant independent of that margin. Since $T^{-\varepsilon_T} = B^{-\varepsilon}$, integration against $|\phi|$ proves (46). This is a pointwise application of the uniform estimate, not an invocation of an unproved varying-epsilon limit. For fixed frequency margin epsilon the displayed $B^{-\varepsilon} \log^4(2T+2)$ tends to zero. $\square$

The class is nonempty: a nonzero smooth bump supported in the square $[3/16, 5/16]^2$ has support in $\Omega_{1/8}$. Functions in this single positive-frequency class are allowed to be complex; no reality, positivity, or full permutation symmetry is assumed. Exchanging indices i_2, i_3 corresponds to exchanging the arguments of F . Simultaneous horizontal reflection negates all deltas and conjugates the kernel denominators; it preserves the full statistic by reindexing, without termwise invariance. All multiplicity and partial-diagonal conventions remain those of the master identity.

For project horizontal coordinates $x_\rho = \delta_\rho \log T$, the first imaginary shift is

$$-L_T(x_{\rho_{i_2}} + x_{\rho_{i_1}})/\log T,$$

and similarly for the second. Thus (45) is a weighted full-zero correlation, not a replacement by F of the real ordinate differences alone. The factor 8 is unchanged, $1/T$ is already in K_T , and no Jacobian or extra L_T^2 appears in the frequency integral.

Fix T, ϕ, ω before removing independent zero cutoffs by the displayed absolute majorant. Only then take $T \rightarrow \infty$, with fixed test function, support margin, and smoothing. Varying families may use the explicit bound, but a vanishing conclusion requires its right side to vanish. The condition $T > 2\pi$ is strict; parameter and support-margin boundaries inside Ω_ε are included.

This theorem covers arbitrary test functions in the stated interior Fourier class with the native kernel retained. Removing that kernel, handling frequency axes or the origin, taking a real-ordinate-only statistic, or proving the full Work Package B theorem requires additional analysis. The zero interior limit establishes none of a GUE main term, a relative asymptotic, or a new horizontal constraint.

13 Boundary layers in the positive frequency quadrant

Task: proof and bookkeeping. The following two claims are UNCONDITIONAL, with the stated support restriction, and have status **proved-draft**. Put

$$Q = [0, \infty)^2, \quad Q_\kappa = \{(\xi, \eta) \in Q : \xi + \eta \leq 1 - \kappa\}, \quad 0 < \kappa < 1.$$

A smooth test on Q means the restriction ψ of a function in $C_c^\infty(\mathbb{R}^2)$. Require its support relative to Q to lie in Q_κ . Its values on the axes are its continuous traces. Write $N_1(\psi) = \|\psi\|_{\infty,Q} + \|\partial_\xi \psi\|_{\infty,Q} + \|\partial_\eta \psi\|_{\infty,Q}$. For the estimates in this section assume $T \geq 2\pi e$, and set

$$b = \log B, \quad B = T/(2\pi), \quad q = \log T, \quad L = \log(2T + 2), \quad \mathcal{W} = 1 + W_\infty + W_1.$$

Thus $b \geq 1$, $q = b + \log(2\pi)$, and $L \asymp q \asymp b$ with absolute constants. In frequency integrals always set $X = B^\xi, Y = B^\eta, Z = XY$. In particular $1 \leq X, Y \leq Z \leq B^{1-\kappa} < T$; no positive lower frequency margin is imposed.

Lemma 9 (TRIPLE-BOUNDARY-ESTIMATES-001). *With G and μ_ω as in Lemma 6, put*

$$R_T(X, Y) = \mathcal{C}_{3,T}(X, Y) - q^3/Z^2 - (q + \mu_\omega)G(X, Y).$$

There is an effective absolute constant such that

$$\int_Q |\psi(\xi, \eta) R_T(B^\xi, B^\eta)| d\xi d\eta \ll \|\psi\|_{\infty,Q} \mathcal{W} (1 + B^{-\kappa} L^4). \quad (47)$$

For fixed admissible smoothing there are also the local profiles

$$q^{-3} \mathcal{C}_{3,T}(e^u, e^v) \longrightarrow e^{-2(u+v)}, \quad (48)$$

$$q^{-2} \mathcal{C}_{3,T}(B^r, e^s) \longrightarrow r(1+s)e^{-2s}. \quad (49)$$

The first convergence is uniform on compact subsets of $u, v \geq 0$; the second on compact subsets of $0 < r < 1, s \geq 0$. The exchanged-axis profile holds as well, including $s = 0$.

Proof. Use the exact budget of Lemma 2, replacing its retained terms by Lemma 6, and use Lemma 4 for the three off-diagonal errors. Consequently R_T is exactly the sum of the eight old non- H errors, all nineteen signed H words, E_D, E_J , and $-R_{\text{res}}$. There is no repeated retained term.

Here is the integrated budget. Each entry bounds the integral of the absolute value over Q_κ , before multiplication by $\|\psi\|_{\infty,Q}$. Absolute effective constants are suppressed. The last row is expanded separately below.

Source	Integrated bound
E_{archcube}	L^2/b^2
E_D	L/b
E_J	$1/T$
$-R_{\text{res}}$	b^{-2}
E_{PPA}	$(1 + W_1)L^3/T$
$E_{PAA}, E_{APA}, E_{AAP}$, each	$(1 + W_1)L^2/T$
$E_{PAP,\text{off}}, E_{APP,\text{off}}, E_{PPP,\text{off}}$, each	$(1 + W_1)B^{-\kappa}L^4$
All nineteen H words	$(1 + W_\infty)(1 + T^{-3/2}L^2)$

For completeness these bounds remain valid at both axes. The archimedean cube bound follows from $L^3 - q^3 \ll L^2$ and $\int_Q e^{-2b(\xi+\eta)} d\xi d\eta = 1/(4b^2)$. For E_D , integrate $L(1+b\eta)e^{-2b\eta}$ in η , and bound the other coordinate's interval by length one; also exchange the coordinates. For G , the substitution $s = b\eta$ gives

$$G_1(B^\xi, B^\eta) d\eta = e^{-2s}(1+s) \left(\xi + \frac{s}{2b} \right) ds, \quad G_1 = \frac{(1 + \log Y) \log(XZ)}{2Y^2}. \quad (50)$$

Integrating with $0 \leq \xi \leq 1$ bounds $\int_{Q_\kappa} G$ absolutely, proving the E_J row. The resonance estimate integrates after $u = b\xi, v = b\eta$ to at most

$$Cb^{-2} \int_0^\infty \int_0^\infty e^{-(u+v)/2} (\log 2 + u + v)^3 du dv \ll b^{-2}.$$

For PPA , its pointwise bound is at most $C(1 + W_1)L^3/(T\sqrt{Z})$. The three single-prime words are at most $C(1 + W_1)L^2/T$, since all scalar denominators only improve their bounds for $X, Y \geq 1$. The mixed off-diagonal bounds use $X/\sqrt{Y}, Y/\sqrt{X} \leq Z, \ell_X, \ell_Y, \ell_Z \ll L$, and $V_1 \ll (1 + W_1)L$. The cubic bound uses $Z/T \leq B^{-\kappa}/(2\pi)$. Integration costs at most unit area.

To account for the remainders, split $H_x = H_x^0 + E_{\text{pole}}(x, \cdot)$, where H^0 is the sum of the archimedean, Dirichlet-series and trivial components. Individually their sup-norm majorants are

$$C_{\text{arch}}/x, \quad C_{\text{DS}}/x, \quad 12x^{-5/2}/(T + 2) \leq 12/(Tx).$$

Thus H_x^0 has majorant C/x . Retain these three component labels under finite distributivity; no cancellation among them is used. For a word with p prime factors, a archimedean main factors and $h \geq 1$ factors H^0 , the uniform product bound is $C(1 + W_\infty)L^{p/2+a}$ times the product of x^{-1} over its nonprime slots. There are at most two primes. A factor in the X, Y, Z slot supplies respectively the decay vectors $(1, 0), (0, 1), (1, 1)$. Any two such vectors have positive sum in both coordinates. Integrating the resulting exponential on $Q_\kappa \subset [0, 1]^2$ gives the complete table

p	a	h	Number	Bound after removing $C(1 + W_\infty)$
2	0	1	3	L/b (the H_Z^0 case has L/b^2)
1	1	1	6	$L^{3/2}/b^2$
0	2	1	3	L^2/b^2
1	0	2	3	$L^{1/2}/b^2$
0	1	2	3	L/b^2
0	0	3	1	b^{-2}

Their sum is bounded absolutely since $b \geq 1, L \asymp b$. Every refined word with at least one pole factor has that factor bounded by $4\sqrt{Z}/T^2 \leq 4T^{-3/2}$. All its remaining nonprime factors are $O(L)$; the prime factors use their weighted L^2 bounds. Its integral is therefore at most $C(1 + W_\infty)T^{-3/2}L^2$. This covers all 208 refined words: $12 \cdot 4 + 6 \cdot 4^2 + 4^3$, grouped without overlap according to whether a pole component occurs. The trivial component's additional T^{-1} may be retained in every occurrence. Summing proves (47); in particular none of these errors can contribute at the integrated scale q .

For the origin profile, on any fixed compact rectangle in u, v , all prime bases and their coefficient norms are bounded. The cube is $q^3 e^{-2(u+v)}$; its evaluation error and all H words are $O_\omega(q^2)$, the mixed retained terms are $O_\omega(q)$, and all other bounds are smaller. This proves (48) uniformly, including $u = v = 0$.

For the axis profile take $r_0 \leq r \leq r_1 < 1, 0 \leq s \leq S$ with $r_0 > 0$. For sufficiently large T , all bases are below T , and $Z/T \ll_S B^{-(1-r_1)}$. The off-diagonal errors are $O_{\omega,S}(B^{-(1-r_1)}L^4)$. The first summand of $(q + \mu_\omega)G/q^2$ is

$$\frac{(q + \mu_\omega)b}{q^2} e^{-2s} (1 + s) \left(r + \frac{s}{2b} \right),$$

which converges uniformly to the stated profile. The other summand and the cube, after division by q^2 , are $O_{\omega,S}(qe^{-2br_0})$. The mixed evaluation errors are $O_{\omega,S}(q + q^2 e^{-2br_0})$ before division; the resonance bound is smaller than q^2 . Among the remainder words, the $H_Y PP$ word is $O_{\omega,S}(q)$,

as are all words with two primes. Every word with fewer primes and a factor in the X or Z slot bounded by an inverse base is $O_{\omega,S}(q^2 e^{-br_0})$; words containing a pole are $O_{\omega}(T^{-3/2} L^2)$. These alternatives cover all remaining H words. The archimedean cube error and the other old errors also become zero uniformly after division by q^2 . This proves (49). Symmetry exchanges X, Y . $\square$

14 The one-sided quadrant limit

Theorem 10 (TRIPLE-QUADRANT-LIMIT-001). *For the preceding tests and smoothing define*

$$\mathcal{B}_T[\psi] = \frac{1}{q} \int_Q \psi(\xi, \eta) \mathcal{C}_{3,T}(B^\xi, B^\eta) d\xi d\eta, \quad \mathcal{B}[\psi] = \frac{1}{4} \psi(0, 0) + \frac{3}{4} \int_0^\infty r \{ \psi(r, 0) + \psi(0, r) \} dr.$$

Uniformly for $T \geq 2\pi e$ and $0 < \kappa < 1$,

$$|\mathcal{B}_T[\psi] - \mathcal{B}[\psi]| \ll N_1(\psi) \mathcal{W}\{b^{-1} + B^{-\kappa} L^3\}, \quad (51)$$

with an effective absolute constant. In particular, for fixed ψ, κ, ω , the limit is $\mathcal{B}[\psi]$.

Define the entire function

$$F_+(z_1, z_2) = \int_Q \psi(\xi, \eta) e^{2\pi i(\xi z_1 + \eta z_2)} d\xi d\eta.$$

With the full-zero arguments of (44), there is also the exact absolutely convergent identity

$$\mathcal{B}_T[\psi] = \frac{8}{q} \sum_{i_1, i_2, i_3 \in \mathcal{I}} K_T(i_2, i_3, i_1) F_+(z_{21}, z_{31}). \quad (52)$$

All multiplicity, horizontal, conjugated-anchor and partial-diagonal conventions of the master identity apply unchanged.

Proof. By Lemma 9, replacing $\mathcal{C}_{3,T}$ by $q^3/Z^2 + (q + \mu_\omega)G$ in $\mathcal{B}_T$ incurs $O(\|\psi\|_\infty \mathcal{W}(b^{-1} + B^{-\kappa} L^3))$. It remains to integrate those two retained terms, not to integrate the local profiles without a dominating bound.

The cube contributes exactly

$$\frac{q^2}{b^2} \int_0^\infty \int_0^\infty e^{-2(u+v)} \psi(u/b, v/b) du dv.$$

The mean value bound $|\psi(u/b, v/b) - \psi(0, 0)| \leq N_1(\psi)(u + v)/b$ holds along line segments in Q . The moments

$$\int_0^\infty s^k e^{-2s} ds = \frac{k!}{2^{k+1}} \quad (k \geq 0) \quad (53)$$

follow by integration by parts, with vanishing exponential boundary terms. Since $q/b = 1 + \log(2\pi)/b$, the cube is $\psi(0, 0)/4 + O(N_1(\psi)/b)$.

For the first mixed term, (50) gives

$$\frac{q + \mu_\omega}{q} \int_0^\infty \int_0^\infty e^{-2s} (1 + s) \left(\xi + \frac{s}{2b} \right) \psi(\xi, s/b) ds d\xi.$$

Both $\psi(\xi, s/b)$ and its trace vanish for $\xi > 1$. On $0 \leq \xi \leq 1$, replacing the former by $\psi(\xi, 0)$ costs at most $N_1(\psi)s/b$; the term $s/(2b)$ also costs $O(N_1(\psi)/b)$ after integration. The coefficient $(q + \mu_\omega)/q$

is $1 + O(1/b)$, uniformly in the normalized nonnegative smoothing. Formula (53) gives $\int_0^\infty (1+s)e^{-2s} ds = 3/4$. Thus this term is $(3/4) \int_0^\infty \xi \psi(\xi, 0) d\xi + O(N_1(\psi)/b)$. Exchanging coordinates gives the other axis. Every majorant used here is integrable, including its first moment. This proves (51).

The compactly supported integrable frequency function $\mathbf{1}_Q \psi$, extended by zero, defines F_+ independently of its values on the boundary, which has planar measure zero. Differentiation under its integral is valid locally uniformly at complex arguments, proving that F_+ is entire. On the real plane it is bounded, hence tempered, and its Fourier transform is $\mathbf{1}_Q \psi$ in the sense of tempered distributions, with the project convention. It need not be Schwartz: a nonzero boundary trace prevents the zero extension from having a continuous representative across that axis, whereas the Fourier transform of a Schwartz function is Schwartz.

At fixed T, ψ, ω , the master majorant gives

$$8 \sum_{i_1, i_2, i_3} \int_Q |\psi K_T(i_2, i_3, i_1) e^{2\pi i(\xi z_{21} + \eta z_{31})}| d\xi d\eta \ll \|\psi\|_{L^1(Q)} B^{1-\kappa} L^3 < \infty.$$

Tonelli/Fubini and the exact phase identity from Theorem 8 now prove (52). This argument uses compact frequency support rather than any real-axis decay of F_+ . Remove independent zero cutoffs at fixed parameters under this majorant; only then let T grow with ψ, κ, ω fixed. $\square$

The limiting functional acts on smooth restrictions to the closed quadrant. Its origin mass $1/4$ and positive-axis densities $3r/4$ are full one-sided masses: no further half-weight convention applies. The traces are determined by the smooth restriction, not by assigning arbitrary values to its zero extension on a null set. The pointwise sizes q^3 at the origin and q^2 on an axis are distinct from the integrated scale q . The mixed terms carry no additional origin atom, as their integrated error estimate shows.

If the test vanishes near both axes, this limiting functional is zero, consistent with the earlier stronger unnormalized interior estimate. Tests near the origin and tests meeting just one axis are both allowed. The axes and the inner boundary $\xi + \eta = 1 - \kappa$ are included; the outer boundary $\xi + \eta = 1$ remains excluded. The identity itself holds for $T > 2\pi$; $T \geq 2\pi e$ is the convenient threshold for the explicit estimate. No symmetry or reality of the test is needed. Swapping the two non-anchor indices exchanges the axis terms; horizontal reflection is still a reindexing of full zeros and kernel denominators. For varying tests, margins or smoothing, a limiting assertion requires the right side of (51) to tend to zero.

This proves a nonzero one-sided weighted correlation limit. The following sections extend across signed frequency sectors to smooth Fourier tests surrounding the origin. Removing the native kernel and extracting horizontal information remain separate tasks.

15 Signed frequency sectors and reciprocal reflection

Task: proof and bookkeeping. The next lemma is **UNCONDITIONAL**; the following theorem also records its explicit Fourier-support restriction. Both have ledger status **proved-draft**. Extend $\mathcal{S}(x, t)$ to every $x > 0$ by the same full-zero series (1), and define $\mathcal{C}_{3,T}(X, Y)$ by the same smoothed product for every $X, Y > 0$. No prime-side formula for $x < 1$ is being imported. For $T > 2\pi$, put

$$\tilde{\mathcal{C}}_T(\xi, \eta) = \mathcal{C}_{3,T}(B^\xi, B^\eta), \quad h(\xi, \eta) = \max\{|\xi|, |\eta|, |\xi + \eta|\}, \quad B = T/(2\pi).$$

Let $R(u, v) = (v, -u - v)$, so $R^3 = I$.

Lemma 11 (TRIPLE-SIGNED-SECTORS-001). *The extended zero series converges absolutely, locally uniformly in $x > 0, t \in \mathbb{R}$, and*

$$\mathcal{S}(x^{-1}, t) = \overline{\mathcal{S}(x, t)}. \quad (54)$$

For all real frequencies,

$$\tilde{\mathcal{C}}_T(R(\xi, \eta)) = \tilde{\mathcal{C}}_T(\xi, \eta), \quad \tilde{\mathcal{C}}_T(-\xi, -\eta) = \overline{\tilde{\mathcal{C}}_T(\xi, \eta)}, \quad \tilde{\mathcal{C}}_T(\eta, \xi) = \tilde{\mathcal{C}}_T(\xi, \eta). \quad (55)$$

The six closed cones $\sigma R^k Q$, $\sigma \in \{1, -1\}$, $k = 0, 1, 2$, cover the plane with disjoint interiors. Their overlaps are on $\xi\eta(\xi + \eta) = 0$, and for $(u, v) \in Q$,

$$\det(\sigma R^k) = 1, \quad h(\sigma R^k(u, v)) = u + v.$$

The master zero-side identity holds for every $X, Y > 0$. In particular, if an integrable ϕ is supported in $h \leq 1 - \kappa$, where $0 < \kappa < 1$, its integrated absolute zero-series majorant is

$$8 \sum_{i_1, i_2, i_3} \int_{\mathbb{R}^2} |\phi(\xi, \eta) K_T(i_2, i_3, i_1) e^{2\pi i(\xi z_{21} + \eta z_{31})}| d\xi d\eta \ll \|\phi\|_1 B^{1-\kappa} \log^3(2T + 2). \quad (56)$$

Here the complex zero arguments are still (44).

Proof. Since $|\delta_j| < 1/2$,

$$\left| \frac{2x^{\delta_j + i(\gamma_j - t)}}{D_j(t)} \right| \leq \frac{8}{3} \max(x, x^{-1})^{1/2} (1 + (t - \gamma_j)^2)^{-1}.$$

On a compact set of x, t , this is a constant times $(1 + \gamma_j^2)^{-1}$, whose zero sum converges by the imported counting bound. This proves local uniform absolute convergence. Conjugate the series and reindex by the multiplicity-preserving map $(\delta, \gamma) \mapsto (-\delta, \gamma)$. Its denominator changes from $\overline{D_j(t)}$ to the denominator of the reflected zero, and the numerator becomes $2x^{-\delta_j - i(\gamma_j - t)}$. This proves (54) with no restriction on the horizontal locations of the zeros.

The frequency integrand is now

$$\mathcal{S}(B^\xi, t) \mathcal{S}(B^\eta, t) \mathcal{S}(B^{-\xi-\eta}, t).$$

Cyclic permutation of its three frequencies gives R ; exchange of the first two factors gives the coordinate swap. Negating all three frequencies conjugates the product, and the smoothing is real. This proves (55).

The six maps and their images are recorded explicitly below. All inequalities in this table include equality; only the interiors are disjoint. A conjugation flag means conjugate $\tilde{\mathcal{C}}_T(u, v)$.

Map	Image of (u, v)	Cone	Conjugate?
I	(u, v)	$\xi, \eta \geq 0$	no
R	$(v, -u - v)$	$\xi \geq 0, \xi + \eta \leq 0$	no
R^2	$(-u - v, u)$	$\eta \geq 0, \xi + \eta \leq 0$	no
$-I$	$(-u, -v)$	$\xi, \eta \leq 0$	yes
$-R$	$(-v, u + v)$	$\xi \leq 0, \xi + \eta \geq 0$	yes
$-R^2$	$(u + v, -u)$	$\eta \leq 0, \xi + \eta \geq 0$	yes

The mixed-sign quadrants split according to the sign of $\xi + \eta$, proving coverage and disjointness of the six interiors. The three positive matrices are

$$I, \quad \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}, \quad \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}.$$

Each matrix and its negative has determinant one. The absolute values of the three frequencies on each cone are $u, v, u + v$, in some order. This proves the formula for h , as well as the global identity

$$h(\xi, \eta) = \frac{|\xi| + |\eta| + |\xi + \eta|}{2}. \quad (57)$$

On a seam the three frequencies are $r, 0, -r$, in some order. The integrand is $\mathcal{S}(1, t)|\mathcal{S}(B^r, t)|^2$, which is real because (54) makes $\mathcal{S}(1, t)$ real. Thus the conjugated and unconjugated descriptions agree on all seams, including the origin.

The product of the three absolutely convergent series has the same phase cancellation and factor eight as before, for arbitrary positive bases. For an individual summand its horizontal numerator is bounded by

$$\left| B^{\xi(\delta_{i_2} + \delta_{i_1}) + \eta(\delta_{i_3} + \delta_{i_1})} \right| \leq B^{(|\xi| + |\eta| + |\xi + \eta|)/2} = B^{h(\xi, \eta)}.$$

It is essential here to collect the coefficient of the anchor delta before taking absolute values. The three denominator majorants and the zero-counting bound then give an integrated absolute sum $O(B^h \log^3(2T + 2))$. Integrating against $|\phi|$ proves (56) and the asserted master extension. All interchanges are at fixed parameters under these majorants. The symmetry identities concern the full occurrence sum; reindexing only some slots need not preserve an individual index-diagonal subseries. $\square$

16 Schwartz tests with hexagonal Fourier support

Let $\mathcal{H} = \{h < 1\}$ and $\mathcal{H}_\kappa = \{h \leq 1 - \kappa\}$, $0 < \kappa < 1$. Every compact subset of $\mathcal{H}$ lies in some $\mathcal{H}_\kappa$. The outer hexagon has vertices $(1, 0), (0, 1), (-1, 1), (-1, 0), (0, -1), (1, -1)$, all excluded.

Theorem 12 (TRIPLE-SIGNED-TEST-FUNCTION-001). *Let $\phi \in C_c^\infty(\mathcal{H})$ be arbitrary and complex-valued, and let F be its entire inverse Fourier transform in the project convention. Then $F|_{\mathbb{R}^2}$ is Schwartz and $\widehat{F} = \phi$. For $T > 2\pi$, define*

$$\begin{aligned} \mathcal{A}_T[\phi] &= \frac{1}{q} \int_{\mathbb{R}^2} \phi(\xi, \eta) \tilde{\mathcal{C}}_T(\xi, \eta) d\xi d\eta, \quad q = \log T, \\ \mathcal{A}[\phi] &= \frac{3}{2} \phi(0, 0) + \frac{3}{2} \int_{\mathbb{R}} |r| \{\phi(r, 0) + \phi(0, r) + \phi(r, -r)\} dr. \end{aligned} \quad (58)$$

There is the exact absolutely convergent identity

$$\mathcal{A}_T[\phi] = \frac{8}{q} \sum_{i_1, i_2, i_3 \in \mathcal{I}} K_T(i_2, i_3, i_1) F(z_{21}, z_{31}). \quad (59)$$

If $\text{supp } \phi \subset \mathcal{H}_\kappa$, then, uniformly for $T \geq 2\pi e$,

$$|\mathcal{A}_T[\phi] - \mathcal{A}[\phi]| \ll \|\phi\|_{C^1} (1 + W_\infty + W_1) \{b^{-1} + B^{-\kappa} \log^3(2T + 2)\}, \quad b = \log B. \quad (60)$$

Here $\|\phi\|_{C^1} = \|\phi\|_\infty + \|\partial_\xi \phi\|_\infty + \|\partial_\eta \phi\|_\infty$. The constant is effective and absolute, including independence from κ . For fixed ϕ, κ, ω , the limit is $\mathcal{A}[\phi]$.

Proof. Compact smooth frequency support gives the entire extension by locally uniform differentiation and gives Schwartz decay on the real plane by repeated integration by parts. Fourier inversion has exactly the negative-forward, positive-inverse 2π convention already fixed. Lemma 11 and its absolute majorant permit Fubini with the zero sums and prove (59).

For $k = 0, 1, 2$, put, on Q ,

$$\psi_{+,k}(u, v) = \phi(R^k(u, v)), \quad \psi_{-,k}(u, v) = \overline{\phi(-R^k(u, v))}.$$

Each is the restriction of a compactly supported smooth function on the plane, with relative support in Q_k . The chain rule gives $N_1(\psi_{\pm,k}) \leq 2\|\phi\|_{C^1}$, since the absolute row sums of every matrix are at most two. The planar seams have measure zero. Change variables by the six determinant-one maps to obtain the exact decomposition

$$\mathcal{A}_T[\phi] = \sum_{k=0}^2 \left\{ \mathcal{B}_T[\psi_{+,k}] + \overline{\mathcal{B}_T[\psi_{-,k}]} \right\}. \quad (61)$$

The conjugation of the test in the negative sectors is necessary for complex tests. Apply Theorem 10 to each term. Its real coefficients ensure that conjugating the negative-sector limit restores $\phi(-R^k(u, v))$, with no conjugate on the resulting values of ϕ . The six errors sum to the bound (60).

Each sector contributes $\phi(0, 0)/4$, giving $3\phi(0, 0)/2$. The images of the two coordinate rays under all six maps are

$$(r, 0), (0, r), (-r, r), (-r, 0), (0, -r), (r, -r), \quad r \geq 0,$$

each occurring twice. One way to check the incidence is to list the columns of $I, R, R^2, -I, -R, -R^2$; they are precisely those six directions with multiplicity two. Thus each ray carries $2(3r/4) dr$. Pairing opposite rays gives (58). In particular the diagonal line is parameterized by $(r, -r)$ with measure dr , not Euclidean arclength. There is no missing square-root-of-two factor. Both adjacent sectors contribute to a limiting ray even though the finite- T planar seam has measure zero. No extra half weights are introduced, and the origin term is counted once per sector.

At fixed T, ϕ, ω , remove independent zero cutoffs using (56). Only then take $T \rightarrow \infty$, with the test, margin and smoothing fixed. The explicit bound remains valid for varying families, but a convergence claim requires its right side to tend to zero. $\square$

A smooth nonzero bump supported in $(-1/4, 1/4)^2$, nonzero at the origin, supplies a test crossing all three seams: its support has $h \leq 1/2$. Thus the class includes real Schwartz tests by choosing $\phi(-\xi, -\eta) = \overline{\phi(\xi, \eta)}$; this symmetry is optional. All tests are allowed to be complex and need no permutation symmetry or positivity. A test supported strictly inside one sector has zero limiting functional, consistent with the quadrant interior result.

The origin and all three internal lines are included. The outer boundary $h = 1$ is excluded; this proof does not extend it by a continuity argument. The normalization is still precisely $8/q$ on the zero side, with $1/T$ already in K_T and no extra mean-spacing Jacobian. Both ordinate signs, full multiplicities, all five index patterns, and the imaginary sums of horizontal displacements remain present. The result supplies a Schwartz test-function theorem for the native weighted full-zero observable. The later comparison lemma identifies its sine benchmark, and the final theorem assembles the result. Kernel removal and horizontal extraction require additional analysis; completion of Work Package B is not asserted here.

17 An explicit full-zero kernel profile

Task: proof and bookkeeping. Assumptions: **UNCONDITIONAL**. The correlation consequence also has the stated support restriction. The next three claims have ledger status **proved-draft**.

The critical-line specialization restricts kernel parameters; it does not impose RH on the unconditional zero sums.

For $K_T(j, k, \ell)$ put

$$a = \gamma_j - \gamma_\ell, \quad b = \gamma_k - \gamma_\ell, \quad z_1 = a - i\delta_j, \quad z_2 = b - i\delta_k, \quad z_3 = i\delta_\ell, \quad d_{mn} = z_m - z_n.$$

Here z_1, z_2, z_3 are integration centers, distinct from the scaled test arguments z_{21}, z_{31} . Allow $|\delta_j|, |\delta_k|, |\delta_\ell| \leq 1/2$ in the analytic profile, including the closed parameter endpoints; actual non-trivial zeros lie in the open strip.

Lemma 13 (TRIPLE-KERNEL-PROFILE-001). *For real a, b and the preceding horizontal parameters, the absolutely convergent integral*

$$\mathcal{J}(a, b; \delta_j, \delta_k, \delta_\ell) := \int_{\mathbb{R}} \prod_{m=1}^3 \frac{1}{1 + (v - z_m)^2} dv \quad (62)$$

has the closed form

$$\mathcal{J} = \pi \frac{24 + d_{12}^2 + d_{13}^2 + d_{23}^2}{(4 + d_{12}^2)(4 + d_{13}^2)(4 + d_{23}^2)}. \quad (63)$$

The formula includes coincident centers. Its denominators never vanish on the stated domain. It is smooth in the real parameters, locally uniformly up to all horizontal endpoints, and satisfies

$$\begin{aligned} \mathcal{J}(a, b; \delta_j, \delta_k, \delta_\ell) &= \mathcal{J}(b, a; \delta_k, \delta_j, \delta_\ell), \\ \mathcal{J}(a, b; -\delta_j, -\delta_k, -\delta_\ell) &= \overline{\mathcal{J}(a, b; \delta_j, \delta_k, \delta_\ell)}. \end{aligned}$$

Proof. Each center has imaginary part at most $1/2$ in absolute value. The real-axis denominator bound used for the master identity gives an integrable product majorant. On compact real-gap sets, this product is $O((1 + v^2)^{-3})$, uniformly in the closed horizontal cube.

First suppose that the three centers are distinct. The upper poles are $z_m + i$, separated from the real axis by at least $1/2$; the lower poles are $z_m - i$. Upper and lower poles cannot coincide, since $|\operatorname{Im} d_{mn}| \leq 1$. Closing a large semicircle in the upper half-plane gives no arc contribution: for fixed centers the integrand is $O(|v|^{-6})$ there. The residue theorem yields

$$\mathcal{J} = \pi \sum_{m=1}^3 \prod_{n \neq m} \frac{1}{d_{mn}(d_{mn} + 2i)}. \quad (64)$$

Here the factor π is $2\pi i/(2i)$.

For an explicit cancellation check write $A = d_{13}, B = d_{23}, D = A - B = d_{12}$. Multiply the rational sum in (64) by $D^2 A^2 B^2 (D^2 + 4)(A^2 + 4)(B^2 + 4)$. Its three terms are

$$\begin{aligned} &D(D - 2i)A(A - 2i)B^2(B^2 + 4), \\ &D(D + 2i)B(B - 2i)A^2(A^2 + 4), \\ &A(A + 2i)B(B + 2i)D^2(D^2 + 4). \end{aligned}$$

Their sum is $D^2A^2B^2(24 + D^2 + A^2 + B^2)$. To verify this polynomial identity, expand the three lines: their imaginary parts, after removing $2iABD$, sum to

$$-(D + A)B(B^2 + 4) + (B - D)A(A^2 + 4) + (A + B)D(D^2 + 4) = 0$$

using $D = A - B$. For the real parts the needed remaining identity is

$$-DAB^2(B^2 + 4) + DBA^2(A^2 + 4) - ABD^2(D^2 + 4) = 3D^2A^2B^2;$$

divide by ABD at generic parameters and substitute $D = A - B$, or multiply out to cover zero parameters. These two identities give the stated sum and hence (63).

For $d = x + iy$, $|y| \leq 1$, $\operatorname{Re}(4 + d^2) = 4 + x^2 - y^2 \geq 3 + x^2$. Thus the final denominators have no zeros, including at coincident centers. Dominated convergence under the compact-set majorant extends the formula to those centers; the residue expression with separate simple poles is not used there. The rational formula also proves the asserted smoothness and effective derivative bounds on compact gap sets. Exchange of the first two factors proves the first symmetry. Conjugation of the integral proves the second, with all three deltas negated. $\square$

18 Freezing the height weight with a summable error

Extend ω by zero to $\mathbb{R}$, as its compact smooth support already permits, and write $W_\infty = \|\omega\|_\infty$, $W'_\infty = \|\omega'\|_\infty$, $L = \log(2T + 2)$. The derivative norm here is a new explicit smoothing dependence, distinct from the earlier $W_1 = \|\omega'\|_1$.

Lemma 14 (TRIPLE-KERNEL-LOCALIZATION-001). *For $T \geq 3$, all zero occurrences satisfy*

$$K_T(j, k, \ell) = \frac{\omega(\gamma_\ell/T)}{T} \mathcal{J}(a, b; \delta_j, \delta_k, \delta_\ell) + E_{\text{kernel}}(j, k, \ell), \quad (65)$$

where, uniformly in all ordinates and horizontal parameters,

$$|E_{\text{kernel}}(j, k, \ell)| \leq \frac{64\pi W'_\infty}{27T^2\{4 + (a - b)^2\}}. \quad (66)$$

More importantly, over the full occurrence multiset,

$$\sum_{j,k,\ell} |E_{\text{kernel}}(j, k, \ell)| \ll \frac{W'_\infty L^4 + W_\infty L^3}{T}. \quad (67)$$

All constants are effective and absolute; all index patterns and both ordinate signs are included.

For the tests of Theorem 12, define

$$\begin{aligned} \mathcal{A}_T^\mathcal{J}[\phi] &:= \frac{8}{Tq} \sum_{i_1, i_2, i_3 \in \mathcal{I}} \omega(\gamma_{i_1}/T) \mathcal{J}(\gamma_{i_2} - \gamma_{i_1}, \gamma_{i_3} - \gamma_{i_1}; \delta_{i_2}, \delta_{i_3}, \delta_{i_1}) \\ &\quad \times F(z_{21}, z_{31}), \quad q = \log T. \end{aligned} \quad (68)$$

This series is absolutely convergent. For support in $h \leq 1 - \kappa$, $0 < \kappa < 1$, and $T \geq 2\pi e$, the additional named error is

$$|\mathcal{A}_T[\phi] - \mathcal{A}_T^\mathcal{J}[\phi]| \ll \|\phi\|_1 (W_\infty + W'_\infty) B^{-\kappa} L^3. \quad (69)$$

Consequently $\mathcal{A}_T^\mathcal{J}$ has the same limit $\mathcal{A}[\phi]$ as in (58), with error bounded by the sum of (60) and (69).

Proof. Substitute $t = \gamma_\ell + v$ in the kernel. Its three denominators become exactly those of (62). Subtract the constant $\omega(\gamma_\ell/T)$ inside the integral. The Lipschitz bound on the difference of weights gives

$$|E_{\text{kernel}}| \leq \frac{64W'_\infty}{27T^2} \int_{\mathbb{R}} \frac{|v| dv}{(1+v^2)(1+(v-a)^2)(1+(v-b)^2)}.$$

Use $|v|/(1+v^2) \leq 1/2$ and

$$\int_{\mathbb{R}} \frac{dv}{(1+(v-a)^2)(1+(v-b)^2)} = \frac{2\pi}{4+(a-b)^2}.$$

For completeness the latter identity follows by the same upper-pole calculation: with $d = a - b \neq 0$, its value is $\pi\{[d(d+2i)]^{-1} + [d(d-2i)]^{-1}\}$. This simplifies to $2\pi/(d^2+4)$, and dominated convergence includes $d = 0$. This proves (66). That tuplewise estimate alone is not summed over all triples.

For the summable bound put $k(v) = (1+v^2)^{-1}$, $Q(t) = \sum_j k(t - \gamma_j) \ll \log(|t| + 2)$, and $\Delta_\ell(t) = \omega(t/T) - \omega(\gamma_\ell/T)$. Tonelli applied to absolute values gives the upper bound

$$\frac{64}{27T} \int_{\mathbb{R}} Q(t)^2 \sum_{\ell} k(t - \gamma_\ell) |\Delta_\ell(t)| dt. \quad (70)$$

This is legitimate as a nonnegative integral before proving it finite. Split the error integrals into three disjoint pieces. The first has $|t - \gamma_\ell| \leq T$. The second has $|t - \gamma_\ell| > T$ and $t \in [T, 2T]$. The third has $|t - \gamma_\ell| > T$, $t \notin [T, 2T]$, and $\gamma_\ell \in [T, 2T]$. Outside these pieces the weight difference is zero. Call them $E_{\text{kernel,near}}$, $E_{\text{kernel,far-center}}$, and $E_{\text{kernel,far-anchor}}$, respectively.

On the near piece a nonzero difference forces $0 \leq t \leq 3T$. The local counting bound, including multiplicities, gives

$$\sum_{|t-\gamma_\ell| \leq T} |t - \gamma_\ell| k(t - \gamma_\ell) \ll L \sum_{0 \leq m \leq \lceil T \rceil} (1+m)^{-1} \ll L^2.$$

Indeed each unit shell contains $O(L)$ occurrences uniformly on this range, and its weight is $O((1+m)^{-1})$. As $|\Delta_\ell(t)| \leq W'_\infty |t - \gamma_\ell|/T$, the integral in (70), restricted to the near piece, is $O(W'_\infty L^4/T)$.

For $t \in [T, 2T]$, distant shells of radius $2^m T$ contain $O(2^m T(L+m))$ occurrences. Hence

$$\sum_{|t-\gamma_\ell| > T} k(t - \gamma_\ell) \ll T^{-1} \sum_{m \geq 0} 2^{-m} (L+m) \ll L/T.$$

The far-center piece uses $|\Delta_\ell| \leq 2W_\infty$, $Q(t)^2 \ll L^2$, and an interval of length T , giving $O(W_\infty L^3/T)$ after the outer factor $1/T$.

For the far-anchor piece, there are $O(TL)$ possible anchor occurrences. At $|v| = |t - \gamma_\ell| > T$ with $\gamma_\ell \in [T, 2T]$, $|t| + 2 \leq 3|v| + 2$, so

$$\int_{|t-\gamma_\ell| > T} Q(t)^2 k(t - \gamma_\ell) dt \ll \int_{|v| > T} \frac{\log^2(3|v| + 2)}{v^2} dv \ll T^{-1} \sum_{m \geq 0} 2^{-m} (L+m)^2 \ll L^2/T.$$

Here $|\Delta_\ell(t)| = |\omega(\gamma_\ell/T)| \leq W_\infty$. Multiplying by the anchor count and the outer $1/T$ gives the same $O(W_\infty L^3/T)$ bound. We have thus retained all sources:

Summed absolute error	Bound
$E_{\text{kernel,near}}$	$O(W'_\infty L^4/T)$
$E_{\text{kernel,far-center}}$	$O(W_\infty L^3/T)$
$E_{\text{kernel,far-anchor}}$	$O(W_\infty L^3/T)$

These arguments use only the reconstructed local counts and kernel bound PAIR-COUNT-001, PAIR-COUNT-KERNEL-001. They prove (67), including all far anchors and negative ordinates. All cutoff removals follow from the finite nonnegative majorants just established.

The original master estimate gives $\sum |K_T| \ll L^3$. The triangle inequality and (67) therefore also give absolute summability of $\omega(\gamma_\ell/T)\mathcal{J}/T$. For the complex test arguments the compact-frequency representation, with the anchor delta collected as in (56), gives uniformly in all zero triples

$$|F(z_{21}, z_{31})| \leq \|\phi\|_1 B^{1-\kappa}.$$

Multiplying the summable error by this bound and $8/q$ proves (69), since $B^{1-\kappa}/T = B^{-\kappa}/(2\pi)$ and $L/q \ll 1$. It proves absolute convergence of the profile statistic as well. The old and new errors have now been checked separately for smoothing uniformity, so their sum is a valid error for the inherited main term. Fix the test, margin and smoothing before the limit; varying families must make both displayed errors tend to zero. $\square$

19 Microscopic profile and critical-line specialization

Corollary 15 (TRIPLE-KERNEL-CRITICAL-SPECIALIZATION-001). *Put*

$$h_{12} = \delta_j - \delta_k, \quad h_{13} = \delta_j + \delta_\ell, \quad h_{23} = \delta_k + \delta_\ell.$$

The zero-gap horizontal profile is

$$\mathcal{J}(0, 0; \boldsymbol{\delta}) = \pi \frac{24 - h_{12}^2 - h_{13}^2 - h_{23}^2}{(4 - h_{12}^2)(4 - h_{13}^2)(4 - h_{23}^2)}. \quad (71)$$

For every fixed $R > 0$, uniformly for $|u|, |v| \leq R$ and $\boldsymbol{\delta} \in [-1/2, 1/2]^3$, as $T \rightarrow \infty$,

$$\mathcal{J}(u/L_T, v/L_T; \boldsymbol{\delta}) = \mathcal{J}(0, 0; \boldsymbol{\delta}) + O_R(L_T^{-1}). \quad (72)$$

For kernel parameters $\boldsymbol{\delta} = \mathbf{0}$, separately,

$$\mathcal{J}(a, b; \mathbf{0}) = \pi \frac{24 + a^2 + b^2 + (a - b)^2}{(4 + a^2)(4 + b^2)(4 + (a - b)^2)}. \quad (73)$$

Its microscopic expansion is

$$\mathcal{J}(u/L_T, v/L_T; \mathbf{0}) = \frac{3\pi}{8} - \frac{5\pi}{32L_T^2}(u^2 + v^2 - uv) + O_R(L_T^{-4}). \quad (74)$$

The constants are effective and uniform in the horizontal cube for (72). In particular a critical-line tuple with these bounded scaled gaps has

$$K_T(j, k, \ell) = \frac{3\pi}{8T} \omega(\gamma_\ell/T) + O_R \left(\frac{W_\infty}{TL_T^2} + \frac{W'_\infty}{T^2} \right).$$

This is a tuplewise assertion on critical-line parameters, not a replacement of every tuple in the unconditional correlation sum.

Proof. At zero gaps, $d_{mn} = -ih_{mn}$; substitute into (63) to get (71). Uniform first derivatives on a compact real-gap set times the closed horizontal cube follow from the same rational formula and $|4 + d_{mn}^2| \geq 3$. The mean value estimate proves (72). It does not require convergence of the horizontal parameters as T varies. The corresponding tuplewise kernel error relative to the zero-gap horizontal profile is $O_R(W_\infty/(TL_T) + W'_\infty/T^2)$.

Set all three deltas to zero to obtain (73). If $s = a^2 + b^2 + (a - b)^2$, its denominator is $64 + 16s + O((|a| + |b|)^4)$. Thus

$$\mathcal{J}(a, b; \mathbf{0})/\pi = \frac{3}{8} - \frac{5}{64}s + O((|a| + |b|)^4).$$

Since $s = 2(a^2 + b^2 - ab)$, this is (74). Uniform Taylor remainder bounds on compact sets follow from nonvanishing denominators. Finally apply (66), whose right side is at most $16\pi W'_\infty/(27T^2)$, to obtain the tuplewise kernel formula. All microscopic limits fix R before $T \rightarrow \infty$. $\square$

The profile (71) still depends on the horizontal parameters. For example, at the algebraic parameters $\delta_j = \delta_k = \delta_\ell = d$, it is $\pi(3 - d^2)/(8(1 - d^2)^2)$, which at $d = 1/4$ is $94\pi/225$, rather than $3\pi/8$. This is a parameter calculation, not an assertion that an off-line zeta zero exists.

The critical-line normalization comparison is the exact identity

$$\frac{8}{T \log T} \frac{3\pi}{8} = \frac{3 \log(T/2\pi)}{2 \log T} \frac{1}{TL_T}. \quad (75)$$

Its coefficient relative to $(TL_T)^{-1}$ tends to $3/2$. This explains that coefficient through the local critical-line kernel; the full main-term comparison is proved in the next section.

Assumption separation. All three claims above are unconditional. Evaluating a kernel at $\delta = \mathbf{0}$, or applying it to a tuple already known to lie on the line, needs no global hypothesis. The statement that this specialization applies to every zero tuple uses RH; that conditional interpretation is not used in any preceding proof. Even under that interpretation, the bounded-gap expansion alone does not justify replacing $\mathcal{J}$ by $3\pi/8$ in the infinite sum: an additional summed error and tail estimate is needed. Unconditionally, real-plane Schwartz decay cannot by itself localize the complex test arguments. The full-gap rational profile and the summable height-localization estimate have been proved for the actual full-zero sum; constant-kernel replacement and horizontal consequences remain separate tasks. The next section identifies its limiting functional.

20 Comparison with the all-ordered sine benchmark

This section is an unconditional deterministic Fourier calculation followed by an application of the already proved weighted correlation theorem. It uses the unit-density sine determinant as a benchmark. For provenance, Conrey–Snaith, *In support of n -correlation*, author manuscript arXiv:1212.5537v2, Theorem 1, equation (2), p. 2, records the distinct-index unitary determinant with kernel $S_N(\alpha) = \sin(N\alpha/2)/\sin(\alpha/2)$. The elementary scaling $N^{-1}S_N(2\pi x/N) \rightarrow \sin(\pi x)/(\pi x)$ fixes our unit-density convention. We prove the benchmark identities below directly; no zeta correlation theorem from that source is imported.

Put $s(x) = \sin(\pi x)/(\pi x)$, with $s(0) = 1$, and define

$$\begin{aligned} R_2(r) &= 1 - s(r)^2, \\ R_3(u, v) &= \det \begin{pmatrix} 1 & s(u) & s(v) \\ s(u) & 1 & s(u-v) \\ s(v) & s(u-v) & 1 \end{pmatrix} \\ &= 1 - s(u)^2 - s(v)^2 - s(u-v)^2 + 2s(u)s(v)s(u-v). \end{aligned} \quad (76)$$

Define the all-ordered benchmark measure M_3 by its pairing with a Schwartz function F :

$$\begin{aligned} \mathcal{S}_3[F] := \langle M_3, F \rangle &= \int_{\mathbb{R}^2} F(u, v) R_3(u, v) du dv + F(0, 0) \\ &+ \int_{\mathbb{R}} R_2(r) \{F(0, r) + F(r, 0) + F(r, r)\} dr. \end{aligned} \quad (77)$$

Every pairing here and below is bilinear, with no complex conjugation. All integrals converge absolutely since the densities are bounded and the restrictions of a Schwartz function to these lines are Schwartz. Line delta distributions mean integration in the displayed coordinate r ; in particular $\delta(u-v)$ pairs as $\int F(r, r) dr$, without an arclength factor. These measures represent the five partitions of three ordered occurrence indices. For zeta, unequal indices can still represent equal complex zeros or equal ordinates; nothing here changes that convention or asserts simplicity of zeta zeros.

Lemma 16 (TRIPLE-SINE-MEASURE-001). *With the project's negative-forward Fourier convention, put*

$$\tau(r) = (1 - |r|)_+, \quad m(r) = 1 - \tau(r), \quad c(\xi, \eta) = (1 - h(\xi, \eta))_+, \quad h = \max(|\xi|, |\eta|, |\xi + \eta|).$$

As tempered distributions,

$$\widehat{M}_3 = \delta_{(0,0)} + m(\xi)\delta(\eta) + m(\eta)\delta(\xi) + m(\xi)\delta(\xi + \eta) + g(\xi, \eta), \quad (78)$$

where the final term is a planar density,

$$g(\xi, \eta) = 1 - \tau(\xi) - \tau(\eta) - \tau(\xi + \eta) + 2c(\xi, \eta). \quad (79)$$

It vanishes on $h \leq 1$. It is the Fourier transform of the reduced ordinary third cumulant distribution

$$\begin{aligned} C_3^{\text{ord}} &= \delta_{(0,0)} - \delta(u)s(v)^2 - \delta(v)s(u)^2 - \delta(u-v)s(u)^2 \\ &+ 2s(u)s(v)s(u-v). \end{aligned} \quad (80)$$

The factorial third cumulant is the last cycle term alone and has transform $2c$; it does not vanish in this region.

Proof. Write $I = [-1/2, 1/2]$. The integral representation $s(x) = \int_I e^{2\pi i x t} dt$ gives $\widehat{s^2} = \mathbf{1}_I * \mathbf{1}_I = \tau$, including the endpoints. For the cycle $s(u)s(v)s(u-v)$, the interval parameters have frequency vector $(a+d, b-d)$. Consequently its transform is

$$\int \mathbf{1}_I(d) \mathbf{1}_I(\xi - d) \mathbf{1}_I(\eta + d) dd = |I \cap (\xi + I) \cap (-\eta + I)| = c(\xi, \eta).$$

Indeed, the span of the three centers $0, \xi, -\eta$ is $\max(|\xi|, |\eta|, |\xi + \eta|)$; the common interval has length $(1 - h)_+$. This is a distributional Fourier calculation on Schwartz tests: the compact parameter integrations can be exchanged with the real-plane integral under the integrable majorant $|F(u, v)|$. The two-factor calculation has the same justification. Lifting the one-dimensional transforms to the plane gives the following complete table. Here a scalar function without a delta is a planar density.

Index partition	Physical measure	Fourier transform
123	$\delta_{(0,0)}$	1
12 3	$\delta(u)R_2(v)$	$\delta(\eta) - \tau(\eta)$
13 2	$\delta(v)R_2(u)$	$\delta(\xi) - \tau(\xi)$
23 1	$\delta(u - v)R_2(u)$	$\delta(\xi + \eta) - \tau(\xi + \eta)$
1 23	$R_3(u, v) du dv$	$\widehat{R}_3$

The determinant expansion gives

$$\widehat{R}_3 = \delta_{(0,0)} - \tau(\xi)\delta(\eta) - \tau(\eta)\delta(\xi) - \tau(\xi)\delta(\xi + \eta) + 2c.$$

For the last line, use $u = r + w, v = w$, a determinant-one change of variables. Thus no $\sqrt{2}$ factor occurs on the frequency line $\xi + \eta = 0$ either. Summing the five rows proves (78). Notice that the frequency atom at the origin comes from the constant physical density in R_3 ; the physical all-equal atom instead transforms to the constant frequency density.

The elementary identity $|\xi| + |\eta| + |\xi + \eta| = 2h$ follows by separating equal and opposite signs of ξ, η . Hence on $h \leq 1$, $g = 1 - (3 - 2h) + 2(1 - h) = 0$, also at the axes and vertices. For example $g(3/4, 3/4) = 1/2$, so this cancellation is support dependent. Finally let $c_2(x) = \delta_0(x) - s(x)^2$, the reduced ordinary covariance of the unit-density benchmark, and lift each one-variable distribution with constant density in the remaining coordinate. Direct subtraction from (77) yields

$$M_3 = 1 + c_2(u) + c_2(v) + c_2(u - v) + C_3^{\text{ord}}.$$

This is the third moment–cumulant identity with mean density one; Fourier transformation gives $\widehat{C}_3^{\text{ord}} = g$. The determinant’s connected factorial cycle has transform $2c$, which equals two at the origin. All equalities are distributional; no infinite zero sum or asymptotic limit enters this lemma. $\square$

Lemma 17 (TRIPLE-MAIN-TERM-COMPARISON-001). *Let $\phi \in C_c^\infty(\mathcal{H})$, $F = \mathcal{F}^{-1}\phi$, and $\text{supp } \phi \subset \{h \leq 1 - \kappa\}$, $0 < \kappa < 1$. No reality or symmetry of ϕ is required. Then*

$$\begin{aligned} \mathcal{S}_3[F] &= \phi(0, 0) + \int_{\mathbb{R}} |r| \{ \phi(r, 0) + \phi(0, r) + \phi(r, -r) \} dr, \\ \mathcal{A}[\phi] &= \tfrac{3}{2} \mathcal{S}_3[F]. \end{aligned} \tag{81}$$

With the previous notation $B = T/(2\pi)$, $b = \log B$, $q = \log T$, $L = \log(2T + 2)$, retain the two errors separately:

$$\begin{aligned} \mathfrak{E}_{\text{signed}} &= \|\phi\|_{C^1} (1 + W_\infty + W_1) \{b^{-1} + B^{-\kappa} L^3\}, \\ \mathfrak{E}_{\text{kernel}} &= \|\phi\|_1 (W_\infty + W'_\infty) B^{-\kappa} L^3. \end{aligned}$$

Under the existing smoothing hypotheses, uniformly for $T \geq 2\pi e$,

$$\begin{aligned} \mathcal{A}_T[\phi] &= \tfrac{3}{2} \mathcal{S}_3[F] + O(\mathfrak{E}_{\text{signed}}), \\ \mathcal{A}_T^{\mathcal{J}}[\phi] &= \tfrac{3}{2} \mathcal{S}_3[F] + O(\mathfrak{E}_{\text{signed}} + \mathfrak{E}_{\text{kernel}}). \end{aligned} \tag{82}$$

Constants are effective and absolute as in the preceding bounds.

Proof. For the stated Fourier convention and bilinear pairing, $\langle M_3, F \rangle = \langle \widehat{M}_3, \phi(-\cdot) \rangle$. Every distribution in (78) is centrally even, so this also equals $\langle \widehat{M}_3, \phi \rangle$, without conjugating ϕ . The planar term vanishes on its support; on each line $m(r) = |r|$ there. This proves the first identity, and comparison with (58) proves the second. Theorem 12 and Lemma 14 now give (82), with no new analytic error. Fix test, support margin and smoothing before taking $T \rightarrow \infty$; varying families must make both displayed errors vanish. $\square$

Normalization, support, and scope. For $F(x_2 - x_1, x_3 - x_1)$, the positive-inverse three-frequency vector is $(-\xi - \eta, \xi, \eta)$, of ℓ^1 -size $2h$. In a negative-exponential representation its sign is reversed. Thus our margin corresponds to $\sum |\lambda_j| \leq 2 - 2\kappa$; this is a convention translation, not an import of a zeta theorem. The benchmark cancellation holds even on $h = 1$, but the zeta estimate still requires compact support in $h < 1$. No boundary extension is claimed.

The finite-height local normalization factor in (75) is $3b/(2q)$, whose limit is $3/2$. It is not set equal to $3/2$ at finite height. Multiplying $\mathcal{A}_T$ or $\mathcal{A}_T^{\mathcal{J}}$ by $2/3$ gives the benchmark as its limiting functional, while retaining every kernel and complex test argument of the original statistic. The result identifies the main term of that unconditional weighted full-zero statistic. It neither removes the kernel nor asserts an unweighted real-ordinate zeta correlation. Likewise the ordinary-cumulant cancellation proved above is a property of the benchmark; transferring it to a weighted zeta cumulant would require compatible lower-order weighted statistics.

21 An unconditional smoothed three-level theorem

Task: proof synthesis and exposition. The following self-contained statement collects the weighted correlation result in unit benchmark normalization. Its ledger status is **proved-draft**; its assumptions are **UNCONDITIONAL** and the stated Fourier-support restriction.

Theorem 18 (TRIPLE-SMOOTHED-CORRELATION-001). *Let $\mathcal{I}$ index all occurrences of nontrivial zeta zeros, with multiplicity and both signs of the ordinate, and write $\rho_j = 1/2 + \delta_j + i\gamma_j$. Let $\omega \in C_c^\infty((1, 2))$ be real and nonnegative with $\int \omega = 1$, extended by zero to $\mathbb{R}$. Fix $0 < \kappa < 1$ and a complex-valued $\phi \in C_c^\infty(\mathbb{R}^2)$ such that*

$$\text{supp } \phi \subset \{(\xi, \eta) : h(\xi, \eta) \leq 1 - \kappa\}, \quad h(\xi, \eta) = \max(|\xi|, |\eta|, |\xi + \eta|).$$

Use the negative-forward 2π Fourier convention and its entire positive-inverse extension

$$F(z_1, z_2) = \int_{\mathbb{R}^2} \phi(\xi, \eta) e^{2\pi i(\xi z_1 + \eta z_2)} d\xi d\eta.$$

Then $F|_{\mathbb{R}^2}$ is Schwartz and $\widehat{F} = \phi$. No reality or permutation symmetry of the test is required.

For $T \geq 2\pi e$, put

$$B = T/(2\pi), \quad b = \log B, \quad q = \log T, \quad L_T = b/(2\pi), \quad L = \log(2T + 2).$$

For a triple (i_1, i_2, i_3) , set

$$a_2 = \gamma_{i_2} - \gamma_{i_1}, \quad a_3 = \gamma_{i_3} - \gamma_{i_1}, \quad z_{21} = L_T\{a_2 - i(\delta_{i_2} + \delta_{i_1})\}, \quad z_{31} = L_T\{a_3 - i(\delta_{i_3} + \delta_{i_1})\}.$$

Define the profile $\mathcal{J}(a_2, a_3; \epsilon_2, \epsilon_3, \epsilon_1)$ for real gaps and $|\epsilon_j| \leq 1/2$ as follows:

$$c_1 = a_2 - i\epsilon_2, \quad c_2 = a_3 - i\epsilon_3, \quad c_3 = i\epsilon_1, \quad d_{mn} = c_m - c_n,$$

$$\mathcal{J}(a_2, a_3; \epsilon_2, \epsilon_3, \epsilon_1) = \pi \frac{24 + d_{12}^2 + d_{13}^2 + d_{23}^2}{(4 + d_{12}^2)(4 + d_{13}^2)(4 + d_{23}^2)}.$$

The c_m are integration centers, not scaled test arguments; the positive sign in c_3 records the conjugated anchor. The rational formula is defined also at coincident centers and horizontal endpoints.

The following sum is absolutely convergent:

$$\mathcal{R}_{3,T}^{\mathcal{J}}[F; \omega] := \frac{16}{3Tq} \sum_{i_1, i_2, i_3 \in \mathcal{I}} \omega(\gamma_{i_1}/T) \mathcal{J}(a_2, a_3; \delta_{i_2}, \delta_{i_3}, \delta_{i_1}) \times F(z_{21}, z_{31}). \quad (83)$$

It includes all ordered occurrence triples and all five index partitions. Equality of indices, complex zeros, and ordinates are distinct conditions. Only the anchor is height localized; the other two slots range over all of $\mathcal{I}$.

Put $s(r) = \sin(\pi r)/(\pi r)$, $s(0) = 1$, and

$$R_2(r) = 1 - s(r)^2, \quad R_3(u, v) = \det \begin{pmatrix} 1 & s(u) & s(v) \\ s(u) & 1 & s(u-v) \\ s(v) & s(u-v) & 1 \end{pmatrix}.$$

The all-ordered sine benchmark is

$$\mathcal{S}_3[F] = \int_{\mathbb{R}^2} F(u, v) R_3(u, v) du dv + \int_{\mathbb{R}} R_2(r) \{F(0, r) + F(r, 0) + F(r, r)\} dr + F(0, 0). \quad (84)$$

Each line uses coordinate measure dr , without an arclength factor. All pairings are bilinear, without complex conjugation. Define

$$W_\infty = \|\omega\|_\infty, \quad W_1 = \|\omega'\|_1, \quad W'_\infty = \|\omega'\|_\infty, \quad \|\phi\|_{C^1} = \|\phi\|_\infty + \|\partial_\xi \phi\|_\infty + \|\partial_\eta \phi\|_\infty,$$

and retain the named bounds

$$\mathfrak{E}_{\text{signed}} = \|\phi\|_{C^1} (1 + W_\infty + W_1) \{b^{-1} + B^{-\kappa} L^3\}, \quad \mathfrak{E}_{\text{kernel}} = \|\phi\|_1 (W_\infty + W'_\infty) B^{-\kappa} L^3. \quad (85)$$

There is an effective absolute constant C , independent of T, κ, ϕ, ω , such that

$$\left| \mathcal{R}_{3,T}^{\mathcal{J}}[F; \omega] - \mathcal{S}_3[F] \right| \leq C(\mathfrak{E}_{\text{signed}} + \mathfrak{E}_{\text{kernel}}). \quad (86)$$

In particular, for fixed test, margin and smoothing, $\mathcal{R}_{3,T}^{\mathcal{J}}[F; \omega] = \mathcal{S}_3[F] + o(1)$. For varying families, the same conclusion about the difference holds whenever both displayed error bounds tend to zero.

Proof. The profile is precisely that of Lemma 13. Compact smooth frequency support gives the entire extension by locally uniform differentiation and real-plane Schwartz decay by integration by parts. For the full complex zero arguments,

$$|F(z_{21}, z_{31})| \leq \|\phi\|_1 B^{1-\kappa},$$

since $|\delta_j| < 1/2$ and $(|\xi| + |\eta| + |\xi + \eta|)/2 = h$. Lemma 14 proves absolute summability of the height-weighted profile and of its product with this test bound. Thus all five index subseries and any independent zero-cutoff limits are justified at fixed T, ϕ, ω , before the height limit.

By the definitions, including the anchor order, the exact normalization is

$$\mathcal{R}_{3,T}^{\mathcal{J}}[F; \omega] = \frac{2}{3} \mathcal{A}_T^{\mathcal{J}}[\phi].$$

Lemma 17 therefore proves (86), retaining the two inherited error bounds separately. Lemma 16 identifies the benchmark with (84), whose integrals converge absolutely by Schwartz decay. No additional sum-limit interchange is needed. Fixing the parameters in the stated order makes both errors vanish, since $b \rightarrow \infty$ and $B^{-\kappa} L^3 \rightarrow 0$. $\square$

Interpretation and remaining scope. The theorem is an additive asymptotic for a weighted full-zero statistic. When $\mathcal{S}_3[F] = 0$, its assertion is still meaningful as $o(1)$; no division by the main term is made. The test class includes the symmetric tests requested in the weighted three-level program. The origin and internal frequency axes are included, while the outer boundary $h = 1$ remains excluded by the positive margin. There is no sharp height cutoff or endpoint half weight. The rational profile and the complex arguments retain all horizontal displacements. The functional-equation reindexings used in the signed sector theorem are inherited unchanged by the exact scalar normalization.

This synthesis neither assumes that the deltas vanish nor removes the kernel. An unweighted real-ordinate theorem, a weighted zeta cumulant identity, and horizontal consequences need their own additional arguments. The assembled theorem is available for the Work Package B audit and reproduction tasks; it does not declare that work package closed.

Name	Exact signed definition	Absolute bound
E_{archcube}	$(J_3 - q^3)/Z^2$	$(L^3 - q^3)/Z^2$
E_{PPA}	$\mathfrak{I}_T(P_X, P_Y, A_Z)$	$b_X b_Y V_1/(ZT \log 4)$
E_{PAA}	$-\mathfrak{I}_T(P_X, A_Y, A_Z)$	$C_1 \sqrt{X} V_2/(T Y Z)$
E_{APA}	$-\mathfrak{I}_T(A_X, P_Y, A_Z)$	$C_1 \sqrt{Y} V_2/(T X Z)$
E_{AAP}	$-\mathfrak{I}_T(A_X, A_Y, P_Z)$	$C_1 \sqrt{Z} V_2/(T X Y)$
$E_{PAP, \text{off}}$	$\mathfrak{I}_T(P_X, A_Y, P_Z) - J_1 D(X, Z)/Y$	$(L/Y)(m_X m_Z + d_{X,Z})$
$E_{APP, \text{off}}$	$\mathfrak{I}_T(A_X, P_Y, P_Z) - J_1 D(Y, Z)/X$	$(L/X)(m_Y m_Z + d_{Y,Z})$
$E_{PPP, \text{off}}$	$-\mathfrak{I}_T(P_X, P_Y, P_Z) + R_{\text{res}}$	$U + R_{\text{res}} \leq U + b_X b_Y b_Z$

Table 3: Named errors from the eight words without H . Each V_r may be replaced by the sharper $\Omega_{r,1}$. The mixed-prime and cubic off-diagonal bounds make no assertion of cancellation.